

Appendix SA - The Ecopath with Ecosim and Ecospace modeling approach

Ecopath

The mass balance over a given time-period considered in Ecopath relies on two master equations. The first one describes the production of each functional group i :

$$B_i \times \left(\frac{P}{B}\right)_i = \sum_{j=1}^N B_j \times \left(\frac{Q}{B}\right)_j \times DC_{ji} + \left(\frac{P}{B}\right)_i \times B_i \times (1 - EE_i) + Y_i + E_i + BA_i \quad (\text{eq. A1})$$

where B_i is the biomass, $(P/B)_i$ the production rate, Y_i the total fisheries yield, E_i the net emigration rate, BA_i the biomass accumulation, and EE_i the ecotrophic efficiency (proportion of the production utilized in the ecosystem). B_j is the biomass of j , one of the N predators of functional group i , which eats functional group i in DC_{ji} proportion in its diet, and whose consumption rate is $(Q/B)_j$. For each functional group i , Y_i and DC_{ji} are inputs parameters, and it is necessary to provide an estimate for three of the following parameters to Ecopath: B_i , $(P/B)_i$, $(Q/B)_i$ and EE_i . The parameter not provided is estimated by Ecopath so that the mass-balanced hypothesis is satisfied (Christensen and Walters, 2004).

The second master equation defines a functional group's productivity with respect to its consumption and maintenance needs:

$$Q_i = P_i + R_i + UA_i \quad (\text{eq. A2})$$

where Q_i is the consumption of functional group i , P_i its production, defined in eq. 1 as $B_i \times \left(\frac{P}{B}\right)_i$, R_i its respiration, and UA_i the unassimilated food. Thus, eq. A2 expresses that consumption is the sum of growth (both somatic and gonadal), metabolic costs and waste products (Winberg, 1956).

Ecosim

Principle and main equations. Ecosim is used to represent temporal changes in the ecosystem of interest in response to trophic interactions, fishing and/or environmental changes over time (Christensen *et al.*, 2008; Christensen and Walters, 2004). Therefore, Ecosim relies on a set of differential equations expressing changes in the biomass of each functional group i over a given time interval dt :

$$\frac{dB_i}{dt} = g_i \times \sum_{j=1}^N Q_{ji} - \sum_{j=1}^N Q_{ij} + I_i - (MO_i + F_i + e_i + i_i) \times B_i \quad (\text{eq. A3})$$

The growth rate of functional group i over time interval dt is given by the difference between the matter produced by functional group i , migrations (emigration and immigration rates e_i and i_i), and losses in functional group i 's biomass due to fishing (fishing mortality rate F_i) and natural mortality due to predation (sum of the consumptions Q_{ij} by functional group i 's predators) or other causes (MO_i mortality rate). The matter produced by functional group i is expressed as the product of food consumed (consumption of all functional group j 's

prey) and growth net efficiency g_i , which is given by the ratio of two EwE parameters: $(Q/B)_i$ and $(P/B)_i$. Consumptions are calculated based on the “foraging arena” theory, which divides prey biomasses into vulnerable and invulnerable compartments, which are, respectively, accessible and not accessible to predators (Ahrens *et al.*, 2012). For a given prey-predator couple (i,j) , the consumption of i by j is estimated as follows (Ahrens *et al.*, 2012; Christensen and Walters, 2004):

$$Q_{i,j} = \frac{a'_{i,j} v_{i,j} B_i P_j T_i T_j \frac{M_{i,j}}{D_j}}{v_{i,j} + v_{i,j} T_i M_{i,j} + a'_{i,j} M_{i,j} B_j \frac{T_j}{D_j}} * S_{t,j}(e_e, t) \quad (\text{eq. A4})$$

where B_j is the biomass of the predator j ; $a'_{i,j}$ is the rate of effective search of predator j for prey i ; T_i and T_j correspond, respectively, to the prey and predator relative feeding time; $M_{i,j}$ is the eventual mediation effect of a third organism on the (i,j) trophic relationship; and D_j is the handling time of the predator j for its prey. The vulnerability coefficient $v_{i,j}$ ranges from 1 to infinity and controls the transfer rate of the prey between the vulnerable and invulnerable compartments. For values of $v_{i,j}$ greater to 2, a large increase in B_j results in a large increase in $Q_{i,j}$. Thus, for values of $v_{i,j}$ greater to 2, the quantity of prey i consumed by predator j is mainly influenced by predator j abundance. We are, then, under “top-down control”. Conversely, when $v_{i,j}$ tends to 1, a large increase in B_j has a lower impact on $Q_{i,j}$; we are, then, under “bottom-up control”.

Hindcasting the past. The value of vulnerability coefficients, initially set to 2 by default as a trade-off between bottom-up and top-down controls, can be estimated through fitting the Ecosim model to observed time-series of biomass or catches. Vulnerability coefficients are estimated so as to minimize the weighted sum of squared deviations between log-scale observed and predicted values (where the weights can be set by the user according to the perceived quality of the observed time series). This hindcasting step can be implemented using a “predator-prey strategy”, where each prey-predator interaction is estimated individually, or using a “predator-only strategy”, where a mean vulnerability value per predator is estimated regardless of the consumed prey. The number of vulnerability coefficients to be estimated is limited by the number of observed time-series provided to Ecosim, and can be chosen either by the modeler or according to a stepwise procedure (Scott *et al.*, 2016) that determines the vulnerability coefficients leading to the best fit to time-series data based on Akaike Information Criterion (Akaike, 1974).

Simulating the impact of fishing and environmental changes. The Ecosim model can be forced or driven using various time-series related to fishing or the environment. Forcing time-series of fishing mortality or fishing effort can be applied to, respectively, functional groups or the fisheries that target these functional groups to simulate the temporal impact of fishing. In particular, fishing mortality time-series can be directly employed in the Ecosim differential equations eq. 3.

Environmental changes can be modeled in two main ways. The first one consists of using observed time-series of primary production (Christensen *et al.*, 2009). In EwE, the production of a primary producer i PP_i is determined from its biomass using:

$$PP_i = \frac{r_i \cdot B_i}{1 + B_i \cdot h_i} * pp_{i,t}^{rel} \quad (\text{eq. A5})$$

where r is the largest possible production/biomass ratio, r_i/h_i is the maximum net primary production when the biomass is not limiting production. $pp_{i,t}^{rel}$ is a time-series of relative primary production that can be used to simulate the impact of observed changes in the production of lower trophic levels, resulting for instance from satellite remote sensing of chlorophyll- a concentration.

A second way of simulating the impact of environmental changes in Ecosim is to model mean species consumption as depending on specific environmental variables. This can be done using one or several functional responses f_e related to environmental variables e_e to express a temporal forcing of the trophic group consumption (Fig. 1). The temporal forcing is realized using a multiplier of the consumption $S_{t,i}(e_e, t)$ (eq. 4) calculated as the product of all $f_{e,i}(e_e, t)$ (Fig A1). For any value taken by an environmental variable e_e at time step t , the functional response f_e provides a value between 0 and 1 modulating the size of the foraging arena. In the more recently published works, the functional response f_e can, for instance, represents a preference function to water temperature (Bentley *et al.*, 2017; Corrales *et al.*, 2017; Serpetti *et al.*, 2017) or oxygen concentration (de Mutsert *et al.*, 2016), hence providing a multiplier of the consumption that vary according to the conditions in the the ecosystem.

If this way of modeling environment affects the whole consumption of the predator, some alternative ways of representing temporal effect of environment can target specific couples of predator-prey interactions. This is performed in EwE using the temporal forcing functions that can modify the consumption on a given prey by simultaneously or independently acting as a multiplier of $v_{i,j}$ or $a'_{i,j}$. The effective search rate $a'_{i,j}$ can be written as the ratio of a search rate $a_{i,j}$ and an area $A_{i,j}$ where foraging is possible. The forcing function thus can act as a multiplier of the former, altering the capacity of the predator to hunt (e.g. *turbidity forcing function*) or act as a multiplier of the former, increasing or decreasing the accessibility of preys (e.g. *macrophytes forcing function* restricting the areas where a predator can forage on its preys). Multiplying the exchange rate $v_{i,j}$, the forcing function can account for changes in preys behavior that would facilitate their catch (e.g. *productive fronts functions* representing the aggregation of zooplankton in favorable areas that would make them more vulnerable to their predator because of easier access to their prey). Dahood *et al.* (2019) notably used environmental multipliers of vulnerability, search rate and arena area to model the impact of sea-ice regimes on a Antarctic ecosystem.

Ecospace

Ecospace is the spatial component of the EwE framework. It is used to predict the spatial distribution of each functional group over a spatial horizontal 2D-grid based on both trophic and habitat considerations (Pauly *et al.*, 2000; Walters, 1999). This is done through the definition of a habitat foraging capacity. For a given functional group j , the habitat foraging capacity of each cell c $C_{c,j}$ is defined according to the values of specific environmental variables e_e in the cell and to functional responses of j to these variables. Therefore, it is necessary to provide Ecospace with maps describing the spatial-temporal patterns of environmental variables and functional responses defined similarly as those employed in Ecosim. The capacity value of each cell, ranging from 0 to 1, is calculated as the product of functional responses resulting from local environmental conditions (see schematic Fig. 1). The ability of a predator to feed in a given grid cell depends on the capacity of this grid cell. This is modeled through the modification of predator j 's consumption in cell c , which is calculated as follows:

$$Q_{c,i,j} = \frac{a_{i,j} * v_{i,j} * B_{c,i} * B_{c,j}}{2 v_{i,j} + a_{i,j} * B_{c,j} / C_{c,j}} \quad (\text{eq. A6})$$

Roughly speaking, Ecospace involves running both Ecopath and Ecosim for all the cells of the grid, which differ from one another in terms of functional group abundances and consumptions. However, the cells are not independent, and they exchange biomass with their adjacent neighbors. A biomass $B_{rc,i}^{out}$ exits each cell at each time-step and is relocated to the left, right, below or above cell:

$$B_{rc,i}^{out} = \sum_{d=1}^4 m_{i,rc \rightarrow d} \cdot B_{rc,i} \quad (\text{eq. A7})$$

$$\text{with } \begin{cases} \sum_{d=1}^4 m_{i,rc \rightarrow d} = m_i^0 \\ m_{i,rc \rightarrow d} = m_{i,d \rightarrow rc} \cdot \frac{B_{d,i}}{B_{rc,i}} \end{cases} \quad \forall d \in [1; 4] \quad (\text{eq. A8})$$

where d represents each of the cardinal directions; $B_{d,i}$ is the biomass of functional group i in cell d ; $m_{i,c \rightarrow d}$ and $m_{i,d \rightarrow c}$ are the movement rates from cell c to cell d and from cell d to cell c , respectively; and m_i^0 is the base dispersal rate of functional group i . The dispersal rate quantifies the ability of functional i to disperse randomly over one year ($\text{km} \cdot \text{y}^{-1}$) and is an input of the Ecospace model. All these specifications entail that movement is less likely towards a less suitable habitat. This adjustment of movement rates between adjacent cells based on their relative biomass allows for maintaining spatial gradients created by habitat foraging capacities.

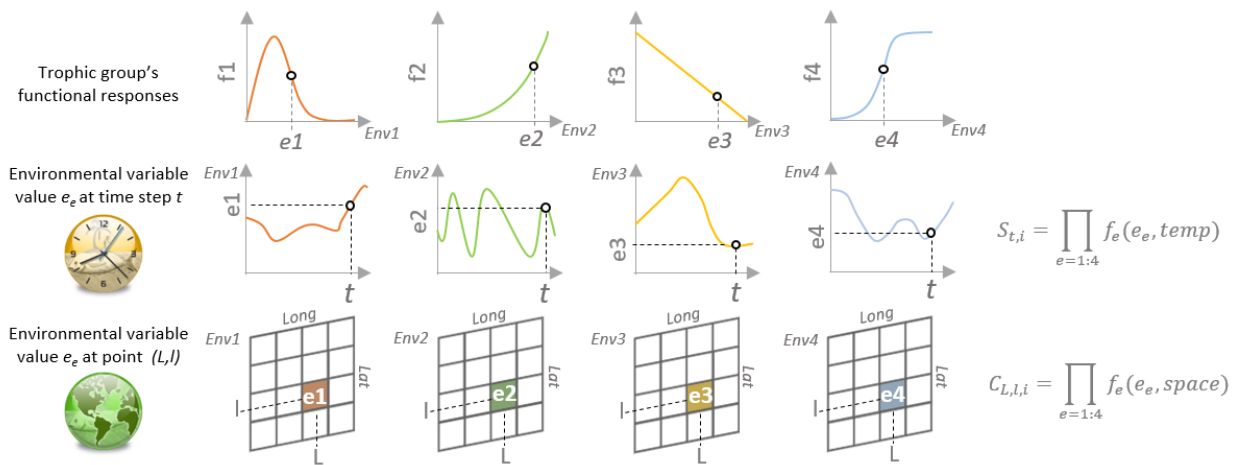


Fig. A1 – Schematic of the integration of environmental effects on species consumption in Ecosim and Ecospace.

Functional responses can be common to the temporal and spatial dimensions.

References SA

- Ahrens, R. N. M., Walters, C. J., and Christensen, V. (2012). Foraging arena theory: Foraging arena theory. *Fish and Fisheries* 13, 41–59. doi:10.1111/j.1467-2979.2011.00432.x.
- Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control* 19, 716–723. doi:10.1109/TAC.1974.1100705.
- Bentley, J. W., Serpetti, N., and Heymans, J. J. (2017). Investigating the potential impacts of ocean warming on the Norwegian and Barents Seas ecosystem using a time-dynamic food-web model. *Ecological Modelling* 360, 94–107. doi:10.1016/j.ecolmodel.2017.07.002.
- Christensen, V., and Walters, C. J. (2004). Ecopath with Ecosim: methods, capabilities and limitations. *Ecological Modelling* 172, 109–139. doi:10.1016/j.ecolmodel.2003.09.003.
- Christensen, V., Walters, C. J., Ahrens, R., Alder, J., Buszowski, J., Christensen, L. B., et al. (2009). Database-driven models of the world's Large Marine Ecosystems. *Ecological Modelling* 220, 1984–1996. doi:10.1016/j.ecolmodel.2009.04.041.
- Christensen, V., Walters, C. J., and Pauly, D. (2008). Ecopath with Ecosim version 6 user guide. Lenfest Ocean Futures Project 2008. Vancouver, Canada: Fisheries Centre, University of British Columbia.
- Corrales, X., Coll, M., Ofir, E., Piroddi, C., Goren, M., Edelist, D., et al. (2017). Hindcasting the dynamics of an Eastern Mediterranean marine ecosystem under the impacts of multiple stressors. *Marine Ecology Progress Series* 580, 17–36. doi:10.3354/meps12271.
- Dahood, A., Watters, G. M., and de Mutsert, K. (2019). Using sea-ice to calibrate a dynamic trophic model for the Western Antarctic Peninsula. *PLoS ONE* 14, e0214814. doi:10.1371/journal.pone.0214814.
- de Mutsert, K., Steenbeek, J., Lewis, K., Buszowski, J., Cowan, J. H., and Christensen, V. (2016). Exploring effects of hypoxia on fish and fisheries in the northern Gulf of Mexico using a dynamic spatially explicit ecosystem model. *Ecological Modelling* 331, 142–150. doi:10.1016/j.ecolmodel.2015.10.013.
- Moullec, F., Gascuel, D., Bentorcha, K., Guénette, S., and Robert, M. (2017). Trophic models: What do we learn about Celtic Sea and Bay of Biscay ecosystems? *Journal of Marine Systems* 172, 104–117. doi:10.1016/j.jmarsys.2017.03.008.
- Pauly, D., Christensen, V., and Walters, C. (2000). Ecopath, Ecosim, and Ecospace as tools for evaluating ecosystem impact of fisheries. *ICES J. Mar. Sci.* 57, 697–706. doi:10.1006/jmsc.2000.0726.
- Scott, E., Serpetti, N., Steenbeek, J., and Heymans, J. J. (2016). A Stepwise Fitting Procedure for automated fitting of Ecopath with Ecosim models. *SoftwareX* 5, 25–30. doi:10.1016/j.softx.2016.02.002.
- Serpetti, N., Baudron, A. R., Burrows, M. T., Payne, B. L., Helaouët, P., Fernandes, P. G., et al. (2017). Impact of ocean warming on sustainable fisheries management informs the Ecosystem Approach to Fisheries. *Scientific Reports* 7. doi:10.1038/s41598-017-13220-7.
- Walters, C. (1999). Ecospace: Prediction of Mesoscale Spatial Patterns in Trophic Relationships of Exploited Ecosystems, with Emphasis on the Impacts of Marine Protected Areas. *Ecosystems* 2, 539–554. doi:10.1007/s100219900101.
- Winberg, G. G. (1956). *Rate of Metabolism and Food Requirements of Fishes: New Information on Metabolic Rate in Fishes*. Fisheries Research Board of Canada.

Appendix SB - The Celtic Sea Ecopath model

Modification of the model structure and input parameters

With the aim of avoiding total modification of the base Celtic Sea EwE model, we limited the structural modifications brought to the pre-existing model. Most of the modifications introduced in the Celtic Sea Ecopath model pertained to the structure of the model. Some of the functional groups that were previously defined based on size or taxonomic criteria were redefined for the present study based on additional (trophic) considerations. Cephalopods were split into benthic and pelagic components to take into account the larger proportion of small pelagic fish in the diet of squids and cuttlefish and the larger proportion of benthic invertebrates in the diet of octopuses. Elasmobranch groups were already separated into large pelagic and demersal components (i.e. “Large pelagic sharks” and “Sharks and Rays”). However, the latter group included a diversity of species with very contrasting feeding habits. For instance, the proportion of fish in the diet of the dogfish (*Squalus acanthias*) was estimated to be greater than 50%, while some ray species like the spotted ray (*Raja montagui*) were almost exclusively benthivorous. Under the assumption that species size largely influences fish metabolic and diet characteristics, the three multi-species demersal fish groups from Moullec *et al.* (2017) 's model were defined based on species size, distinguishing between (≤ 30 cm), medium (≥ 30 cm and ≤ 60 cm) and large demersal fishes (≥ 60 cm). For ecological reasons, we redistributed the species from the three multi-species demersal fish groups into five new functional groups: “Piscivorous demersal fish”, “Epibenthivorous demersal fish”, “Endobenthivorous demersal fish”, “Small benthivorous demersal fish” and “Suprabenthivorous demersal fish”. This new definition of demersal fish groups allows one to distinguish between (i) large species that feed mainly on fish, (ii) medium to large species that eventually feed on fish, but essentially eat large benthic invertebrates that are easily accessible on the bottom (e.g. crawling crustaceans) or invertebrates swimming in the close water layer (e.g. shrimps), (iii) medium to large species that feed on invertebrates inhabiting burrows (e.g. worms or small bivalves), (iv) small benthivorous fish whose diet is similar to the diet of endobenthivorous or epibenthivorous fish, but are of smaller size and hence represents primary prey for many larger demersal fish, and (v) small fish that also represent an important prey source for other demersal fish, that largely feed on plankton, and that potentially move over a larger portion of water column (e.g. that undertake large diel migrations). Finally, based on respect to recent trophic studies in the Celtic Sea (Day *et al.*, 2019; Issac *et al.*, 2017), the multi-stanza cod and hake groups were maintained and a new multi-stanza anglerfish group was created. The list of species included in the model is presented in Tab. 4.B1.

The Celtic Sea diet matrix from Moullec *et al.* (2017) 's model was built based only on literature studies (Bentorcha *et al.*, 2016). We updated the literature search from Moullec *et al.* (2017); see Appendix 3D Tab.3 E1. The updated literature information was then used as informative prior knowledge on the food-web

topology and dietary proportions of all the functional groups in the Bayesian integrated model EcoDiet (Hervann *et al.*, in press; Théro *et al.*, 2020). EcoDiet incorporated stable isotope and stomach content analyses conducted from fish collected recently in the Celtic Sea (within the EATME project, IFREMER; Appendix 3.E). EcoDiet outputs provided new information on diets in the Celtic Sea, especially for commercial species for which both isotope and stomach data were available. The construction of the new diet matrix and the changes introduced in the diet matrix compared to Moullec *et al.* (2017) and their impacts on the estimated Celtic Sea food-web are analyzed in detail in Kopp *et al.* (in prep.). Since the diet matrix constructed from EcoDiet outputs corresponds to 2016, the diet matrix for 1985 was derived from it by correcting dietary proportions to the pro-rata of changes in prey abundance. The diet matrix is presented in Tab. B2

After modification of the diet matrix, in Tab. B2, integration of fisheries catches and discards, in Tab. B3, and slight update of the model parameters, the model was balanced. The biomass and vital rates did not vary greatly regarding the previous version of the model. The modifications required for balancing mainly resulted from diet modifications. The input and post-balancing estimates of the model parameters are presented in Tab. B4. Depending on whether the parameters were estimated by the model or provided as input data, and in the latter case, according to the origin and reliability of the data, a different value of Pedigree scores was attributed. Pedigree scores are presented in Tab. B5.

Table B1 – Functional groups represented in the updated Celtic Sea Ecopath with Ecosim (EwE) model and their species composition. Species included in the model are species that have been either collected during fisheries-independent surveys conducted over the study area or mentioned in official catch statistics. Because of possible taxonomic misidentifications in official catch reports, species reported in official catch statistics were included in the EwE model only if their spatial distribution reported in aquamaps.org included our study area. Biblio, SCA and SIA columns indicate whether species data was integrated in EcoDiet to estimate the diet of the functional group it pertains. Biblio refers to diet knowledge from the literature, SCA to stomach content analyzes and SIA to stable isotope analyzes. A*, B* and C* mean that prior knowledge on the Celtic Sea food-web integrated into EcoDiet, respectively comes from Valentina et al. (2014), comes from Moullec et al. (2015), or was adapted from Ainsworth (2001), Moullec et al. (2017) and Agnetta et al. (2019).

Common name	Latin name	Ecopath group	Comments	Stat.	evhoe	biblio	SCA	SIA
Northern gannet	<i>Sula bassana</i>	Plunge and pursuit divers seabirds				A*		
Common murre	<i>Uria aalge</i>							
Razorbills	<i>Alca torda</i>							
Atlantic puffin	<i>Fratercula arctica</i>							
Herring gull	<i>Larus argentatus</i>	Surface feeders seabirds	They mainly differ from plunge and pursuit divers by their higher consumption of discards					
Yellow-legged gull	<i>Larus michachellis</i>							
Lesser black-backed gull	<i>Larus fuscus</i>							
great black-backed gull	<i>Larus maritimus</i>							
Kittiwakes	<i>Rissa tridactyla</i>							
Minke whale	<i>Balaenoptera acutorostrata</i>	Baleen whales						
Humpback whale	<i>Megaptera novaengliae</i>							
Sei whale	<i>Balaenoptera borealis</i>							
Fin whale	<i>Balaenoptera physalus</i>							
Blue whale	<i>Balaenoptera musculus</i>							
Harbour porpoise	<i>Phocoena phocoena</i>	Toothed whales and Seals				B*		
Common dolphin	<i>Delphinus delphis</i>							
Striped dolphin	<i>Stenella coeruleoalba</i>							
Bottlenose dolphin	<i>Tursiops truncatus</i>							
Risso's dolphin	<i>Grampus griseus</i>							
White-beaked dolphin	<i>Lagenorhynchus albirostris</i>							
Atlantic White-sided dolphin	<i>Lagenorhynchus acutus</i>							
False Killer Whale	<i>Pseudorca crassidens</i>							
Long-finned Pilot Whale	<i>Globicephala melas</i>							
	<i>Globicephala macrorhynchus</i>							
Short-finned Pilot Whale	<i>Orcinus orca</i>							
Killer Whale (Orca)	<i>Orcinus orca</i>							
Halichoerus grypus	<i>Halichoerus grypus</i>							
Phoca vitulina	<i>Phoca vitulina</i>							
Gervais beaked whale	<i>Mesoplodon europaeus</i>							
True's beaked whale	<i>Mesoplodon mirus</i>							
pygmy sperm whale	<i>Kogia breviceps</i>							
dwarf sperm whale	<i>Kogia simus</i>							
Thresher	<i>Alopias vulpinus</i>	Pelagic sharks		x		x		x
Bluntnose sixgill shark	<i>Hexanchus griseus</i>			x	x			
Shortfin mako	<i>Isurus oxyrinchus</i>			x				
Porbeagle	<i>Lamna nasus</i>			x	x	x		
Blue shark	<i>Prionace glauca</i>			x		x		x

Supplementary Material to: The Celtic Sea Through Time and Space: Ecosystem Modeling to Unravel Fishing and Climate Change Impacts on Food-Web Structure and Dynamics
Hervnann P.-Y., Gascuel D., Grüss A., Druon J.-N., Kopp D., Perez I., Piroddi C. and Robert M.

Tope shark	Galeorhinus galeus			x	x	x		x
Shagreen ray	Leucoraja fullonica	Piscivorous demersal elasmobranchs	Demersal sharks and rays that feed largely on fish, especially pelagic species	x	x	x		x
Blue skate	Raja batis			x	x	x		x
White skate	Raja alba			x				x
Angelshark	Squatina squatina			x				
Longnose spurdog	Squalus blainville			x				
Marbled electric ray	Torpedo marmorata			x	x	x		
Picked dogfish	Squalus acanthias			x	x	x		x
Starry smooth-hound	Mustelus asterias	Benthivorous demersal elasmobranchs	Demersal sharks and rays that mainly feed on benthic invertebrates and, to a lesser extent, on small benthic fish	x	x	x		x
Cuckoo ray	Leucoraja naevus			x	x	x		x
Smooth-hound	Mustelus mustelus			x	x	x		x
Blonde ray	Raja brachyura			x	x	x		x
Spotted ray	Raja montagui			x	x	x		x
Thornback ray	Raja clavata			x	x	x		x
Blackmouth catshark	Galeus melastomus			x	x	x		x
Common stingray	Dasyatis pastinaca			x	x			
Common eagle ray	Myliobatis aquila			x	x			
Sandy ray	Raja circularis			x				x
Longnosed skate	Raja oxyrinchus			x				x
Undulate ray	Raja undulata			x	x			
Small-spotted catshark	Scyliorhinus canicula			x	x	x		x
Nursehound	Scyliorhinus stellaris			x	x	x		x
Small-eyed ray	Raja microocellata			x	x			x
Blackbellied angler	Lophius budegassa	Anglerfish		x	x	x	x	x
Angler(=Monk)	Lophius piscatorius			x	x	x	x	x
European seabass	Dicentrarchus labrax	Sea bass		x	x	x		x
European hake	Merluccius merluccius	Hake		x	x	x	x	x
Atlantic cod	Gadus morhua	Cod		x	x	x	x	x
Haddock	Melanogrammus aeglefinus	Haddock		x	x	x	x	x
Whiting	Merlangius merlangus	Whiting		x	x	x	x	x
Megrim	Lepidorhombus whiffiagonis	Megrim		x	x	x	x	x
Norway pout	Trisopterus esmarkii	Pouts		x	x	x		x
Pouting(=Bib)	Trisopterus luscus			x	x	x		x
Poor cod	Trisopterus minutus			x	x	x		x
European plaice	Pleuronectes platessa	Plaice		x	x	x	x	x
Common sole	Solea solea	Sole		x	x	x	x	x
European conger	Conger conger	Piscivorous demersal fish	Demersal fish mainly that feed on fish, both benthic and pelagic. Usually of large size (max. length > 60cm).	x	x	x		x
Pollack	Pollachius pollachius			x	x	x		x
Turbot	Psetta maxima			x	x	x		x
Brill	Scophthalmus rhombus			x	x	x		
Ling	Molva molva			x	x	x		x
John dory	Zeus faber			x	x	x		x
Common lingue	Molva macrophthalma				x	x		x
Common dentex	Dentex dentex			x				
Meagre	Argyrosomus regius			x	x			
Saithe(=Pollock)	Pollachius virens			x		x		x
Atlantic halibut	Hippoglossus hippoglossus			x				x
Silver scabbardfish	Lepidopus caudatus			x	x			x
Blackbelly rosefish	Helicolenus dactylopterus					x		x
Wreckfish	Polyprion americanus			x				
European eel	Anguilla anguilla			x	x			
Atlantic salmon	Salmo salar			x				

Sea trout	Salmo trutta			x				
Black Sea brill	Psetta maeotica				x			x
Atlantic thornyhead	Trachyscorpia cristulata			x				
Barracudas nei	Sphyræna spp			x				
Spotted seabass	Dicentrarchus punctatus				x			
Four-spot megrim	Lepidorhombus boscii				x	x		x
Greater forkbeard	Phycis blennoides			x	x	x		x
Fourbeard rockling	Enchelyopus cimbrius				x	x		x
Red gurnard	Chelidonichthys cuculus			x	x	x		x
Surmullet	Mullus surmuletus			x	x	x		x
Blackspot seabream	Pagellus bogaraveo			x	x	x		x
Argentine	Argentina sphyraena			x	x	x		x
Three-bearded rocling	Gaidropsarus vulgaris				x	x		x
Tub gurnard	Chelidonichthys lucerna			x	x	x		x
Grey gurnard	Eutrigla gurnardus			x	x	x		x
Red scorpionfish	Scorpaena scrofa			x	x			
Black scorpionfish	Scorpaena porcus			x				
Comber	Serranus cabrilla			x				
Greater weever	Trachinus draco			x	x	x		x
Atlantic wolffish	Anarhichas lupus			x				
Gilthead seabream	Sparus aurata			x	x			x
Streaked gurnard	Chelidonichthys lastoviza			x	x			
Thicklip grey mullet	Chelon labrosus			x				
Ballan wrasse	Labrus bergylta			x				
Axillary seabream	Pagellus acarne			x	x	x		x
Common pandora	Pagellus erythrinus			x	x	x		
Black seabream	Spondyliosoma cantharus			x	x			
Golden grey mullet	Liza aurata			x	x			
Thinlip grey mullet	Liza ramada			x	x			
Flathead grey mullet	Mugil cephalus			x				
Piper gurnard	Eutrigla lyra			x	x			x
Lumpfish(=Lumpsucker)	Cyclopterus lumpus			x				
Grey triggerfish	Balistes capriscus				x			
Longfin gurnard	Chelidonichthys obscurus				x			
Lesser weever	Echiichthys vipera				x			
Lesser weever	Labrus mixtus				x			
Amer. Plaice	Hippoglossoides platessoides			x	x	x		x
Cuckoo wrasse	Pagrus pagrus				x			
Common dab	Limanda limanda			x	x	x		x
Witch flounder	Glyptocephalus cynoglossus			x	x	x		x
Lemon sole	Microstomus kitt			x	x	x		x
Thickback sole	Microchirus variegatus			x	x	x		x
Sand sole	Solea lascaris			x	x			x
European flounder	Platichthys flesus			x				x
Dragonet	Callionymus lyra				x	x		x
Spotted ragonet	Callionymus maculatus				x	x		x
Fivebeard rockling	Ciliata mustela				x			x
Sand goby	Pomatoschistus minutus				x	x		x
Butterfly blenny	Blennius ocellaris				x	x		x
Imperial scaldfish	Arnoglossus imperialis				x	x		x
Scale-rayed wrasse	Acantholabrus palloni				x			
Mediterranean scaldfish	Arnoglossus laterna				x	x		x

Supplementary Material to: The Celtic Sea Through Time and Space: Ecosystem Modeling to Unravel Fishing and Climate Change Impacts on Food-Web Structure and Dynamics
Hervnann P.-Y., Gascuel D., Grüss A., Druon J.-N., Kopp D., Perez I., Piroddi C. and Robert M.

Solenette	Buglossidium luteum				x			
Fries's goby	Lesueurigobius friesii				x			
Greater pipefish	Syngnathus acus				x			
Black goby	Gobius niger				x	x		x
Norwegian topknot	Phrynorhombus norvegicus				x			
Lozano's goby	Pomatoschistus lozanoi				x	x		
Red bandfish	Cepola macrophthalma				x	x		x
Silvery pout	Gadiculus argenteus				x	x		x
Silvery lightfish	Maurollicus muelleri				x	x		x
Bogue	Boops boops			x	x			
Greater argentine	Argentina silus			x		x		x
Greater sand-eel	Hyperoplus immaculatus				x			x
Great sandeel	Hyperoplus lanceolatus				x	x		
Small sandeel	Ammodytes tobianus				x	x		
Transparent goby	Aphia minuta				x			x
Smooth sandeel	Gymnammodytes semisquamatus				x			
Longspine snipefish	Macrorhamphosus scolopax				x	x		
Blue whiting	Micromesistius poutassou	Blue whiting		x	x	x	x	x
Boarfish	Capros aper	Boarfish		x	x	x		x
Atlantic mackerel	Scomber scombrus	Mackerel		x	x	x		x
Atlantic horse mackerel	Trachurus trachurus	Horse mackerel		x	x	x		x
Atlantic herring	Clupea harengus	Herring		x	x	x		x
European sprat	Sprattus sprattus	Sprat		x	x	x		x
European pilchard	Sardina pilchardus	Sardine		x	x	x		x
Garfish	Belone belone			x		x		
Atlantic bonito	Sarda sarda			x	x	x		
Golden redfish	Sebastes marinus			x				
Albacore	Thunnus alalunga			x				
Atlantic bluefin tuna	Thunnus thynnus			x		x		
Swordfish	Xiphias gladius			x		x		
Atlantic pomfret	Brama brama			x				
Little tunny	Euthynnus alletteratus			x				
Skipjack tuna	Katsuwonus pelamis			x				
Allis and twaite shads	Alosa alosa, A. fallax			x	x	x		
European anchovy	Engraulis encrasicolus			x	x	x		x
European smelt	Osmerus eperlanus			x				
Atlantic saury	Scomberesox saurus			x				
Sand smelt	Atherina presbyter				x	x		
Atlantic chub mackerel	Scomber colias				x	x		
Chub mackerel	Scomber japonicus				x	x		
Mediterranean horse mackerel	Trachurus mediterraneus				x	x		
Broadtail shortfin squid	Illex coindetii			x	x	x		x
Northern shortfin squid	Illex illecebrosus			x				
European squid	Loligo vulgaris			x	x	x		x
European flying squid	Todarodes sagittatus			x				
European common squid	Alloteuthis subulata				x	x		x
Veined squid	Loligo forbesi				x	x		x
lesser flying squid	Todaropsis eblanae				x	x		x
Common cuttlefish	Sepia officinalis			x	x	x		x
Pink cuttlefish	Sepia orbignyana			x	x	x		x
Elegant cuttlefish	Sepia elegans				x			
Bobtail squid	Sepiola sp				x	x		x

Horned octopus	Eledone cirrhosa			x	x	x		x
Common octopus	Octopus vulgaris			x				
Stout bobtail	Rossia macrosoma				x	x		x
Norway lobster	Nephrops norvegicus	Norway lobster		x	x	x		x
Edible crab	Cancer pagurus	Commercial large crustaceans		x		x		x
European lobster	Homarus gammarus			x	x	x		
Spinous spider crab	Maja squinado/brachydactyla			x	x	x		x
Velvet swimcrab	Necora puber			x	x			
Common spiny lobster	Palinurus elephas			x	x	x		
Shrimps (Sh)			Benthic Shrimps-like invertebrates	x	x	C*		x
Commercial bivalves (ComBiv)			Bivalve suspension feeders that are commercially exploited (e.g. Pectinids)	x	x			x
Suprabenthos (Supra)			Suspension feeders (e.g.mysids) that live in the water layer close to the seabed and with sufficient swimming ability to move within this layer		x			x
Suspension and surface deposit feeders (SSDF)			Suspension feeder filtering suspended particles from the seabed plus depositivores feeding on particles at the sediment-water interface (e.g., polychaetes, reg. urchins, amphipods, isopods, cumaceans)	x	x			x
Subsurface deposit feeders (SubSDF)			Depositivorous organisms that feed on particles within the sediments, by ingesting and filtering sediment or by directly selecting particles (mainly polychaetes, few amphipods and irregular urchins)		x			x
Carnivores and necrophages (CarnNec)			Benthic invertebrates that are either predators or scavengers (small crabs, squat lobsters, welks, ophiurids, worms, amphipods...)	x	x			x
Benthic meiofauna (Meio)			Invertebrates living in the sediment (nematods)		x			
Macrozooplankton (MacroZ)			Zooplankton of size ≥ 1cm (omnivorous and carnivorous; they mainly include euphausiids, large carnivorous plankton, gelatinous, tunicates, thecostomates, chaetognathes, and larvae)					
Mesozooplankton - Large (MesOL)			Zooplankton of size ≥ 2mm and < 1cm. (larger copepods, with limited planktivorous behaviour)					x
Mesozooplankton - Small (MesoS)			Zooplankton of size ≥ 0.5mm and < 2mm (essentially herbivorous copepods)					
Microzooplankton (MicroZ)			Zooplankton of size < 0.5mm (ciliates and heterotrophic flagellates)					
Phytoplankton - Large (PhL)			Phytoplankton of size ≥ 5 µm. Mainly diatoms and dinoflagellates					
Phytoplankton - Small (PhS)			Phytoplankton of size < 5 µm. Nano- and picoplankton					

Hervann P.-Y., Gascuel D., Grüss A., Druon J.-N., Kopp D., Perez I., Piroddi C. and Robert M.

Table B2 – Diet Matrix of the Celtic Sea EwE model - 1985

Prey \ predator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Seabirds - Divers																									
Seabirds - Surface feeders																									
Baleen whales																									
Toothed cetaceans / Seals																									
Pelagic sharks				0.014	0.001																				
Carnivorous dem. elasmobranchs				0.002	0.007																				
Benthivorous dem. elasmobranchs				0.003		0.001																			
Sea bass								0.022																	
Anglerfish adult				0.029	0.009				0.001		0.006		0.002		0.001			0.007			0.001				
Anglerfish juvenile									0.000		0.000	0.001	0.001		0.000			0.000			0.001				
Hake adult				0.004	0.001				0.005		0.019							0.000			0.023				
Hake juvenile								0.000	0.001	0.002	0.010	0.002	0.002	0.002	0.001			0.001			0.001				
Cod adult				0.001	0.006	0.001			0.008						0.002						0.004				
Cod juvenile					0.005	0.000			0.003	0.002					0.001			0.002			0.002	0.001			
Whiting				0.000	0.002	0.000	0.004	0.001	0.026	0.005	0.001	0.002	0.008	0.002	0.003			0.000			0.021	0.002			
Haddock					0.008	0.008	0.007		0.005	0.002			0.005	0.004	0.007			0.001			0.003	0.002			
Pouts				0.080	0.160	0.155	0.126	0.114	0.266	0.241	0.097	0.022	0.153	0.056	0.232	0.002	0.016	0.116			0.142	0.027			
Megrim									0.029	0.006	0.004		0.012	0.003				0.008							
Sole								0.012	0.003					0.010	0.005										
Plaice				0.001		0.001		0.006	0.001				0.010	0.002				0.001			0.000				
Piscivorous demersal fish				0.016	0.005			0.002	0.041				0.001								0.030				
Epibenthivorous demersal fish	0.067			0.023	0.044	0.004	0.005	0.005	0.067	0.010	0.007	0.015	0.006	0.042	0.004		0.016	0.045			0.101				
Endobenthivorous demersal fish			0.104		0.093	0.013		0.009	0.047	0.024			0.001	0.006				0.021			0.002	0.002			
Suprabenthivorous demersal fish							0.000	0.006	0.008	0.017	0.010	0.023	0.004	0.010	0.015	0.001		0.053			0.042			0.027	
Small benthivorous demersal fish			0.065	0.056	0.002	0.017	0.017	0.005	0.015	0.275	0.007	0.003	0.009	0.024	0.005	0.001	0.006	0.073			0.098	0.082	0.002		0.052

	Prey \ predator	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	
1	Seabirds - Divers																										
2	Seabirds - Surface feeders																										
3	Baleen whales																										
4	Toothed cetaceans / Seals																										
5	Pelagic sharks								0.009																		
6	Carnivorous dem. elasmobranchs																										
7	Benthivorous dem. elasmobranchs																										
8	Sea bass																										
9	Anglerfish adult																										
10	Anglerfish juvenile										0.000																
11	Hake adult																										
12	Hake juvenile		0.001								0.000	0.000															
13	Cod adult																										
14	Cod juvenile		0.001								0.000																
15	Whiting		0.006																								
16	Haddock																										
17	Pouts								0.002		0.121																
18	Megrim																										
19	Sole											0.000															
20	Plaice																										
21	Piscivorous demersal fish										0.005																
22	Epibenthivorous demersal fish		0.002						0.002		0.052	0.000															
23	Endobenthivorous demersal fish								0.014																		
24	Suprabenthivorous demersal fish		0.044			0.047			0.005		0.210	0.004															
25	Small benthivorous demersal fish					0.005			0.013		0.013	0.239		0.001													
26	Mackerel								0.004		0.008	0.001															
27	Horse mackerel								0.114		0.083	0.004															
28	Boarfish								0.016																		
29	Sprat		0.001						0.008		0.020																
30	Blue whiting	0.052	0.018			0.026					0.024	0.004															
31	Pilchard								0.010																		
32	Herring		0.008						0.012		0.015																
33	Pelagic - Large								0.016																		
34	Pelagic - Medium	0.004	0.006			0.005		0.002	0.008		0.005																
35	Squids	0.004	0.001			0.008			0.064		0.040	0.004															
36	Benthic cephalopods	0.002	0.009			0.007		0.002	0.039	0.004	0.055	0.053		0.031													
37	Commercial crustaceans											0.036	0.024														
38	Nephrops					0.003					0.004	0.003															
39	Commercial bivalves												0.031				0.003										
40	Shrimps	0.015	0.057	0.003	0.083	0.010		0.013	0.011		0.052	0.090	0.196	0.021		0.016											
41	Carnivores/Necrophages		0.008	0.037	0.051	0.016		0.004	0.040	0.021	0.002	0.302	0.310	0.267		0.183	0.060										
42	Suspension/Surface detritus		0.001	0.032	0.113	0.018	0.093	0.021	0.004	0.077		0.049	0.132	0.213		0.227	0.337										
43	Feeders																										
43	Subsurface deposit feeders			0.004		0.002				0.014		0.012	0.181	0.235		0.115	0.192										
44	Suprabenthos	0.010	0.004	0.051		0.041				0.094		0.160		0.043		0.040						0.104					
45	Benthic meiofauna													0.032		0.040			0.319	0.034							
46	Macrozooplankton	0.521	0.340	0.049	0.014	0.724	0.107	0.236	0.109	0.023	0.036	0.040	0.069	0.053		0.031											
47	Mesozooplankton - Large	0.270	0.369	0.606	0.177	0.066	0.350	0.508		0.252	0.191					0.011						0.110					
48	Mesozooplankton - Small	0.123	0.123	0.218	0.562	0.022	0.447	0.214		0.515	0.064	0.000				0.019				0.038		0.245	0.111				
49	Microzooplankton															0.121				0.113		0.125	0.167	0.075	0.048		
50	Bacteria																	0.053	0.130	0.150	0.091				0.095		
51	Phytoplankton - Large		0.000				0.003							0.048	0.150	0.011	0.012	0.531	0.001	0.206		0.125	0.417	0.283	0.143		
52	Phytoplankton - Small														0.300			0.106		0.206		0.125	0.231	0.566	0.381	0.050	
53	Discards												0.003	0.002			0.001										
54	Detritus												0.056	0.053	0.550	0.186	0.396	0.309	0.549	0.253	0.909	0.166	0.074	0.075	0.333	0.950	
54	Import								0.502																		

Table B3 – EwE inputs 1985 – Fishery landings and discards per functional group

Group name	Landings	Discards
<i>Seabirds - Divers</i>	0	0
<i>Seabirds - Surface feeders</i>	0	0
<i>Baleen whales</i>	0	0
<i>Toothed cetaceans / Seals</i>	0	0
<i>Pelagic sharks</i>	0,001655468	0,002375133
<i>Carnivorous dem. elasmobranchs</i>	0,02998101	0,01303352
<i>Benthivorous dem. elasmobranchs</i>	0,05040251	0,02191128
<i>Sea bass</i>	0,006176509	4,41E-05
<i>Anglerfish</i>		
<i>Anglerfish adult</i>	0,08526047	0,002013557
<i>Anglerfish juvenile</i>	0,0107485	0,008173936
<i>Hake</i>		
<i>Hake adult</i>	0,07081173	0,004222286
<i>Hake juvenile</i>	0,002125019	0,000745109
<i>Cod</i>		
<i>Cod adult</i>	0,0307071	0,000732107
<i>Cod juvenile</i>	0,002948705	0,00483317
<i>Whiting</i>	0,04397539	0,01997533
<i>Haddock</i>	0,01634743	0,007265786
<i>Pouts</i>	0,005402181	0,002315221
<i>Megrim</i>	0,0636155	0,01029495
<i>Sole</i>	0,01406094	0,000316553
<i>Plaice</i>	0,0188	0,01001616
<i>Piscivorous demersal fish</i>	0,07579874	0,03295134
<i>Epibenthivorous demersal fish</i>	0,01493525	0,006492905
<i>Endobenthivorous demersal fish</i>	0,01882821	0,008185319
<i>Suprabenthivorous demersal fish</i>	0,000415831	0,000178213
<i>Small benthivorous demersal fish</i>	4,24E-08	0,004634042
<i>Mackerel</i>	0,0953	0,023825
<i>Horse mackerel</i>	0,1744716	0,001762339
<i>Boarfish</i>	1,00E-08	4,63E-09
<i>Sprat</i>	0,02946571	0
<i>Blue whiting</i>	0,0249508	0,0124754
<i>Pilchard</i>	0,02706771	0
<i>Herring</i>	0,08287889	0,001657578
<i>Pelagic - Large</i>	0,0547801	0
<i>Pelagic - Medium</i>	0,000874821	0,00038036
<i>Squids</i>	0,004737154	0,002059381
<i>Benthic cephalopods</i>	0,01204677	0,005237087
<i>Commercial crustaceans</i>	0,06842601	0,02974603
<i>Nephrops</i>	0,01944769	0,004120504
<i>Commercial bivalves</i>	0,06309443	0
<i>Shrimps</i>	0,000460528	0,000197369
<i>Carnivores/Necrophages</i>	0,02322321	0,009952805
<i>Suspension/Surface detritus Feeders</i>	0,008835248	0,003786535
<i>Subsurface deposit feeders</i>	0	0
<i>Suprabenthos</i>	0	0
<i>Benthic meiofauna</i>	0	0
<i>Macrozooplankton</i>	0	0
<i>Mesozooplankton - Large</i>	0	0
<i>Mesozooplankton - Small</i>	0	0
<i>Microzooplankton</i>	0	0
<i>Bacteria</i>	0	0
<i>Phytoplankton - Large</i>	0	0
<i>Phytoplankton - Small</i>	0	0
<i>Discards</i>	0	0
<i>Detritus</i>	0	0
<i>Sum</i>	1,253057	0,2559104

Table B4 - Ecopath model for 1985 - basic parameters and estimates (in bold) inputs

	Group name	TL	B (t/km ²)	P/B (y ⁻¹)	C/B (y ⁻¹)	EE	P/C (y ⁻¹)	BA (y ⁻¹)	Unassim
1	Seabirds - Divers	4,40	0,00	0,15	135,26	0,00	0,001	0	0.2
2	Seabirds - Surface feeders	3,86	0,00	0,25	114,47	0,00	0,001	0	0.2
3	Baleen whales	3,63	0,07	0,04	4,77	0,00	0,01	0	0.2
4	Toothed cetaceans / Seals	4,88	0,03	0,14	12,00	0,00	0,01	0	0.2
5	Pelagic sharks	4,80	0,12	0,20	1,96	0,60	0,10	0	0.2
6	Piscivorous dem. elasmobranchs	4,39	0,15	0,47	3,13	0,65	0,15	0	0.2
7	Benthivorous dem. elasmobranchs	3,96	0,22	0,52	3,48	0,65	0,15	0	0.2
8	Sea bass	4,18	0,05	0,28	1,52	0,58	0,18	0	0.2
9	Anglerfish adult	4,65	0,26		2,38	0,57	0,31	0	0.2
10	Anglerfish juvenile	4,50	0,04		6,89	0,38	0,19	0	0.2
11	Hake adult	4,49	0,10		2,52	0,96	0,47	0	0.2
12	Hake juvenile	4,00	0,05		5,56	0,22	0,24	0	0.2
13	Cod adult	4,15	0,08		2,58	0,85	0,27	0	0.2
14	Cod juvenile	3,95	0,02		5,04	0,90	0,19	0	0.2
15	Whiting	4,24	0,17	0,92	3,70	0,97	0,25	0	0.2
16	Haddock	3,43	0,06	0,86	3,33	0,98	0,26	0	0.2
17	Pouts	3,61	0,94	1,35	5,40	0,95	0,25	0	0.2
18	Megrim	4,17	0,30	0,45	2,24		0,2	0	0.2
19	Sole	3,32	0,05	0,47	2,58	0,96	0,18	0	0.2
20	Plaice	3,32	0,04	0,81	4,50	0,97	0,18	-0,01	0.2
21	Piscivorous dem. fish	4,45	0,33	0,64	3,21	0,85	0,2	0	0.2
22	Epibenthivorous dem. fish	3,63	0,50	0,86	4,30	0,95	0,2	0	0.2
23	Endobenthivorous dem. fish	3,29	0,15	1,09	5,45	0,95	0,2	0	0.2
24	Suprabenthivorous dem. fish	3,44	0,49	1,88	7,52	0,95	0,25	0	0.25
25	Small benthivorous dem. fish	3,50	0,95	1,83	7,34	0,95	0,25	0	0.25
26	Mackerel	3,58	0,95	0,26	1,04	0,97	0,25	0	0.2
27	Horse mackerel	3,59	6,40	0,25	1,00	0,79	0,25	0	0.2
28	Boarfish	3,30	0,15	1,20	4,80	0,07	0,25	0	0.2
29	Sprat	3,25	0,26	0,85	7,00	0,95	0,12	0	0.2
30	Blue whiting	3,73	0,48	1,43	5,70	0,94	0,25	0	0.2
31	Pilchard	3,22	0,31	0,68	6,80	0,62	0,10	0	0.2
32	Herring	3,37	0,81	0,51	4,59	0,97	0,11	0	0.2
33	Pelagic - Large	4,34	0,20	0,42	2,80	0,90	0,15	0	0.2
34	Pelagic - Medium	3,21	0,16	0,88	5,93	0,95	0,148	0	0.2
35	Squids	4,20	0,09	3,90	13,00	0,9	0,3	0	0.2
36	Benthic cephalopods	3,97	0,23	3,90	13,00	0,9	0,3	0	0.2
37	Commercial crustaceans	3,56	0,78	0,55	3,67	0,80	0,15	0	0.2
38	Nephrops	3,36	0,22	0,60	4,00	0,77	0,15	0	0.2
39	Commercial bivalves	2,00	0,30	3,50	17,50	0,73	0,2	0	0.2
40	Shrimps	3,09	1,20	3,00	20,00	0,95	0,15	0	0.2
41	Carnivores/Necrophages	2,75	5,02	6,11	40,73	0,95	0,15	0	0.2
42	Suspension/Surface detritus Feeders	2,05	15,55	6,35	31,75	0,95	0,2	0	0.2
43	Subsurface deposit feeders	2,48	21,69	2,88	14,40	0,9	0,2	0	0.3
44	Suprabenthos	2,36	1,53	7,70	38,50	0,95	0,2	0	0.2
45	Benthic meiofauna	2,09	11,41	10,00	50,00	0,95	0,2	0	0.4
46	Macrozooplankton	2,70	2,25	7,50	25,00	0,74	0,28	0	0.3
47	Mesozooplankton - Large	2,31	1,98	8,01	26,69	0,90	0,28	0	0.3
48	Mesozooplankton - Small	2,09	9,61	25,35	84,50	0,12	0,28	0	0.3
49	Microzooplankton	2,15	2,52	63,90	213,00	0,70	0,29	0	0.32
50	Bacteria	2,00	1,75	157,44	393,60	0,65	0,31	0	0.32
51	Phytoplankton - Large	1,00	14,19	76,45		0,57		0	0
52	Phytoplankton - Small	1,00	10,49	76,45		0,98		0	0
53	Discards	1,00	0,25			0,62		0,40	0.2
54	Detritus	1,00	124,00			0,79		3,88	0

Assessment of the reliability of the Ecopath model input parameters.

In order to quantify the reliability of the information used as Ecopath input parameters, a Pedigree score was attributed to each of them (Christensen *et al.*, 2004; Morissette, 2007). The criteria used to assign Pedigree scores depend on the type of parameters and are listed in Tab. B5. These Pedigree scores are used to assess the sensitivity of the Ecosim model to Ecopath input parameters (Heymans *et al.*, 2016; Steenbeek *et al.*, 2018; see main article). This sensitivity analysis is conducted based on confidence intervals deduced from the Pedigree scores and indicated in Tab. B5. The green values correspond to high Pedigree scores, hence small uncertainties around parameter values and small confidence intervals. On the contrary, low Pedigree scores in red correspond to very uncertain data, hence large confidence intervals.

Table B5 – (Below) Confidence intervals associated to each basis Ecopath parameter, i.e. biomass B , productivity rate P/B , consumption rate Q/B , diet proportions DC and catches Y , and derived from the

Pedigree. The criteria for employed to set a Pedigree are listed to the right of the table. These confidence intervals are used in the Monte Carlo analysis.

Supplementary Material to: The Celtic Sea Through Time and Space: Ecosystem Modeling to Unravel Fishing and Climate Change Impacts on Food-Web Structure and Dynamics
Hervnann P.-Y., Gascuel D., Grüss A., Druon J.-N., Kopp D., Perez I., Piroddi C. and Robert M.

Functional group	B	P/B	Q/B	DC	Y
Seabirds - Divers	0.3	0.7	0.7	0.8	—
Seabirds - Surface feeders	0.3	0.7	0.7	0.8	—
Baleen whales	0.5	0.7	0.7	0.8	—
Toothed cetaceans / Seals	0.5	0.7	0.7	0.8	—
Pelagic sharks	0.8	0.8	0.5	0.5	0.3
Piscivorous dem. elasmobranchs	0.8	0.8	0.5	0.5	0.3
Benthivorous dem. elasmobranchs	0.8	0.8	0.5	0.3	0.3
Sea bass	0.1	0.5	0.5	0.5	0.3
Anglerfish adult	0.3	0.5	0.5	0.1	0.3
Anglerfish juvenile	0.3	0.7	0.8	0.1	0.3
Hake adult	0.3	0.5	0.5	0.1	0.3
Hake juvenile	0.3	0.7	0.8	0.1	0.3
Cod adult	0.1	0.5	0.5	0.1	0.3
Cod juvenile	0.1	0.7	0.8	0.1	0.3
Whiting	0.1	0.5	0.5	0.1	0.3
Haddock	0.1	0.5	0.5	0.1	0.3
Pouts	0.8	0.8	0.5	0.5	0.3
Megrim	0.3	0.5	0.8	0.1	0.3
Sole	0.1	0.5	0.8	0.1	0.3
Plaice	0.1	0.5	0.8	0.1	0.3
Piscivorous demersal fish	0.8	0.8	0.5	0.5	0.3
Epibenthivorous demersal fish	0.8	0.8	0.5	0.3	0.3
Endobenthivorous demersal fish	0.8	0.8	0.5	0.3	0.3
Suprabenthivorous demersal fish	0.8	0.8	0.5	0.5	0.3
Small benthivorous demersal fish	0.8	0.8	0.5	0.5	0.3
Mackerel	0.5	0.5	0.8	0.3	0.3
Horse mackerel	0.5	0.5	0.8	0.3	0.3
Boarfish	0.5	0.8	0.5	0.3	0.3
Sprat	0.8	0.5	0.5	0.5	0.3
Blue whiting	0.5	0.8	0.5	0.1	0.3
Pilchard	0.5	0.5	0.5	0.5	0.3
Herring	0.1	0.5	0.5	0.5	0.3
Pelagic - Large	0.8	0.8	0.5	0.5	0.3
Pelagic - Medium	0.8	0.8	0.5	0.5	0.3
Squids	0.8	0.8	0.8	0.3	0.3
Benthic cephalopods	0.8	0.8	0.8	0.3	0.3
Commercial crustaceans	0.8	0.5	0.8	0.5	0.3
Nephrops	0.5	0.5	0.8	0.3	0.3
Commercial bivalves	0.8	0.5	0.8	0.5	0.3
Shrimps	0.8	0.5	0.8	0.5	0.3
Carnivores/Necrophages	0.8	0.5	0.8	0.5	0.3
Suspension/Surface detritus Feeders	0.8	0.5	0.8	0.5	0.3
Subsurface deposit feeders	0.8	0.5	0.8	0.5	0.3
Suprabenthos	0.8	0.5	0.8	0.5	—
Benthic meiofauna	0.8	0.5	0.8	0.8	—
Macrozooplankton	0.5	0.8	0.5	0.8	—
Mesozooplankton - Large	0.5	0.8	0.5	0.8	—
Mesozooplankton - Small	0.5	0.8	0.5	0.8	—
Microzooplankton	0.5	0.8	0.5	0.8	—
Bacteria	0.8	0.5	0.8	0.8	—
Phytoplankton - Large	0.3	0.5	—	—	—
Phytoplankton - Small	0.3	0.5	—	—	—
Discards	0.5	—	—	—	—
Detritus	0.8	—	—	—	—

Criteria for Pedigree attribution

Biomass

- 0.1 WGCSE - local stock assessment
- ICES WGs -widely distributed stock assessment wider distribution
- 0.3
- 0.5 Approximate / indirect method
- 0.8 EwE estimate / guesstimate / empirical relationship

P/B

- 0.4 Experimental study
- 0.5 Empirical relationship
- 0.7 Guesstimate
- 0.8 EwE estimate

Q/B

- 0.4 Experimental study
- 0.5 Empirical relationship
- 0.7 Guesstimate
- 0.8 EwE estimate

Diet

- 0.1 EcoDiet integrating SCA, SIA and literature
- 0.3 EcoDiet integrating SIA (representative + large sample size) and literature
- 0.5 EcoDiet integrating SIA (low representativity + small sample size) and literature
- 0.8 EcoDiet integrating literature

Catch

- 0.3 ICES Statlant data / Stock assessments

Assessment of the ecological quality of the Celtic Sea model using a PreBal diagnostic

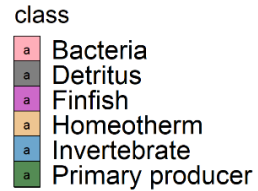
A PreBal diagnostic of the Celtic Sea Ecopath model was conducted in order to assess its ecological quality. The PreBal procedure (Link, 2010) uses general ecological principles to study across functional groups the patterns in the biomass and vital rates or various parameters to assess the relevance of their values.

The changes operated to build the new version of the Celtic Sea Ecopath model from the version inherited from Moullec et al. (2017) did not cause major balancing problem as it had been already balanced and checked using PreBal. Thus, the PREBAL diagnostic was here mainly used to a posteriori validate the quality of the Celtic Sea model.

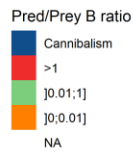
Similarly to Bentley et al. (2019), we listed the rules of thumb implied by PreBal in the following Tab. B6 and indicated whether the required criteria were met and, if not, we discussed about the results. The Figures B1 to 4.B8 illustrate the behavior of the Celtic Sea model regarding the PreBal criteria.

Table B6 - Summary of the criteria required to be validated in PREBAL and corresponding scores in the Celtic Sea.

<i>Criteria</i>	<i>Celtic Sea model results</i>	<i>Comment</i>
Biomass should span 5-7 orders of magnitude	Spans 6 orders of magnitude	–
Biomass slope (log scale) around 5-10% decline with increasing TL	9.3%	–
Are any functional group biomasses notably above/below the line?	Met; Fig. 4.B1	Below B: - The two seabird: B values issued from the literature; lower biomass compared with neighboring areas have been well described; potential anthropogenic impacts - Discards but results only regulated by fishing; Above: - Horse mackerel: large biomass but corresponds to exceptional recruitments combined to low fishing in 1985; - Benthic invertebrate groups: B estimated by the model and support predation of multiple compartments including themselves
Compared across taxa, the ratio between predator and prey biomass should be <1 and ~1-2 decimal places, depending on TL	$71.3\% \subseteq]0.01;1]$ $8.7\% \subseteq]0;0.01]$ $19.8\% > 1$ Fig. 4.B2	The large proportion of cells with a ratio > 1 is related to very is explained by various elements : (i) the integration of prey in the diet even when they represent really low contributions, especially for species whose biomass substantially increased over the study period (e.g. boarfish); (ii) the exceptional biomass levels of particular fish groups that dominate in the ecosystem (e.g. horse mackerel); (iii) the presence of juvenile multi-stanza that are predated by numerous groups and are often in lower abundance.
PB should decline with increasing TL (excluding homeotherms)	Met; Fig. 4.B3	Low values for whales; high values for cephalopods
QB should decline with increasing TL (excluding homeotherms)	Met; Fig. 4.B4	Q/B especially high for seabirds
No taxa should have PB greater than phytoplankton	Met; Fig. 4.B3	–
PQ should fall below 1 for all functional groups	Met; Fig. 4.B5	true for all groups except for homeotherms with really low values
PR should fall below 1 for all functional groups	Met; Fig. 4.B6	–
EE should fall below 1 for all functional groups	Met; Tab. 4.B4	–
Total consumption an production should decrease with increasing TL	Met; Fig. 4.B7 and Fig. 4.B8	–



Celtic Sea Ecopath model and the log10 of biomass.



identifying the trophic relationships due to cannibalism.

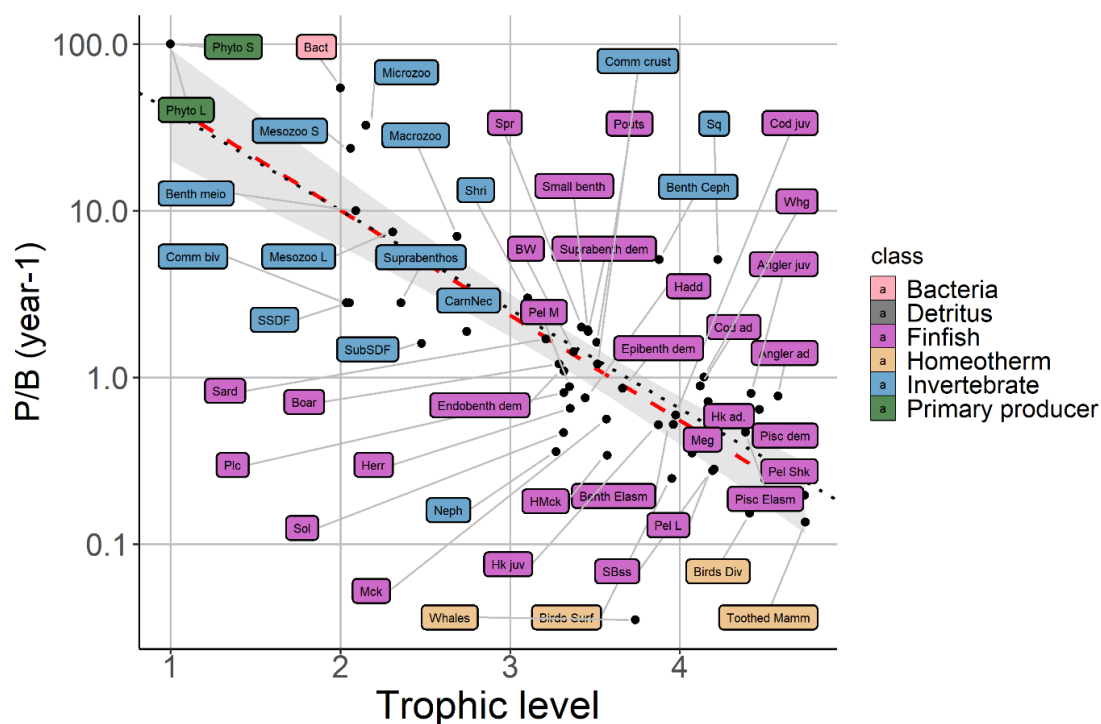


Figure B3 - Negative relationship between the PB rate ($\log_{10}$) and the trophic level of functional groups in the Celtic Sea Ecopath model. The red line corresponds to the regression including homeotherms and the black one excluding homeotherms.

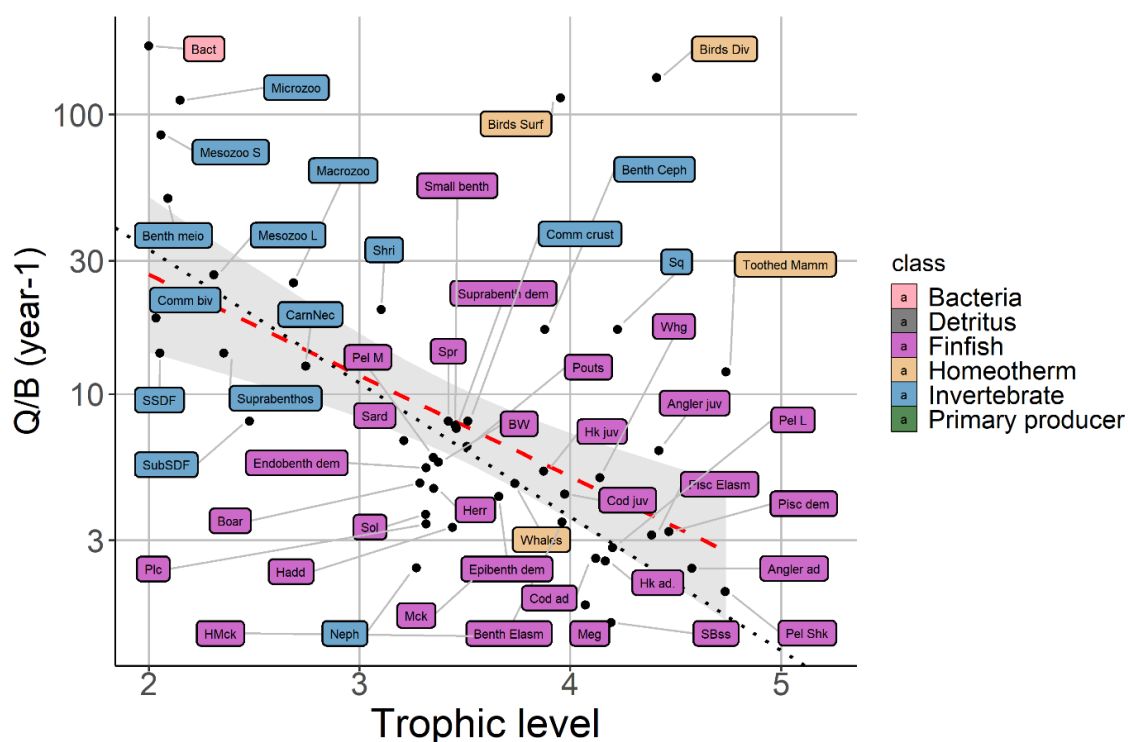


Figure B4 - Negative relationship between the QB rate ($\log$ scale) and the trophic level of functional groups in the Celtic Sea Ecopath model. The red line corresponds to the regression including homeotherms and the black one excluding homeotherms.

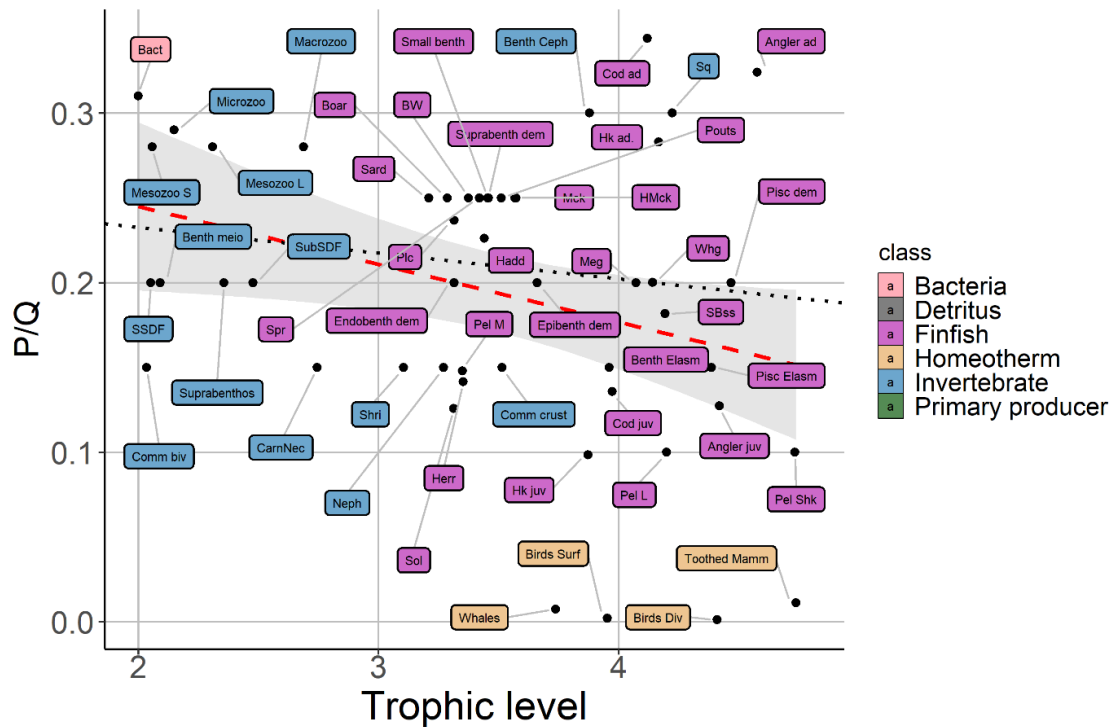


Figure B5 - Negative relationship between the PQ rate (log scale) and the trophic level of functional groups in the Celtic Sea Ecopath model. The red line corresponds to the regression including homeotherms and the black one excluding homeotherms.

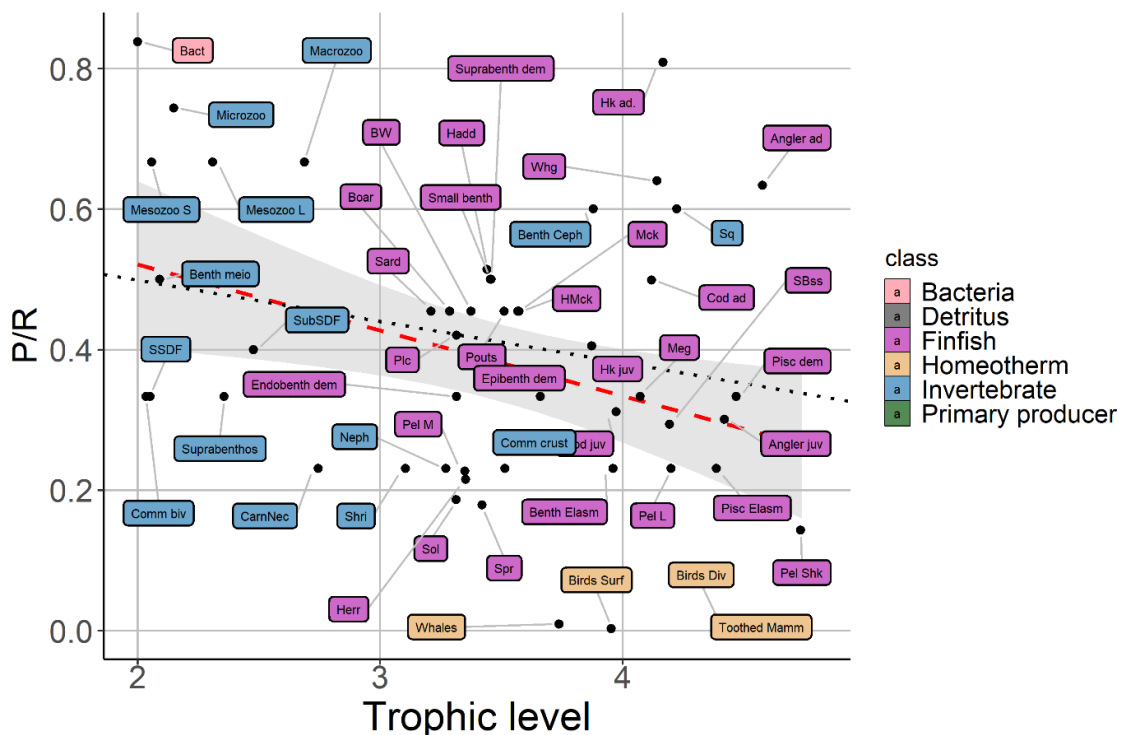


Figure B6 - Negative relationship between the PR rate (log scale) and the trophic level of functional groups in the Celtic Sea Ecopath model. The red line corresponds to the regression including homeotherms and the black one excluding homeotherms.

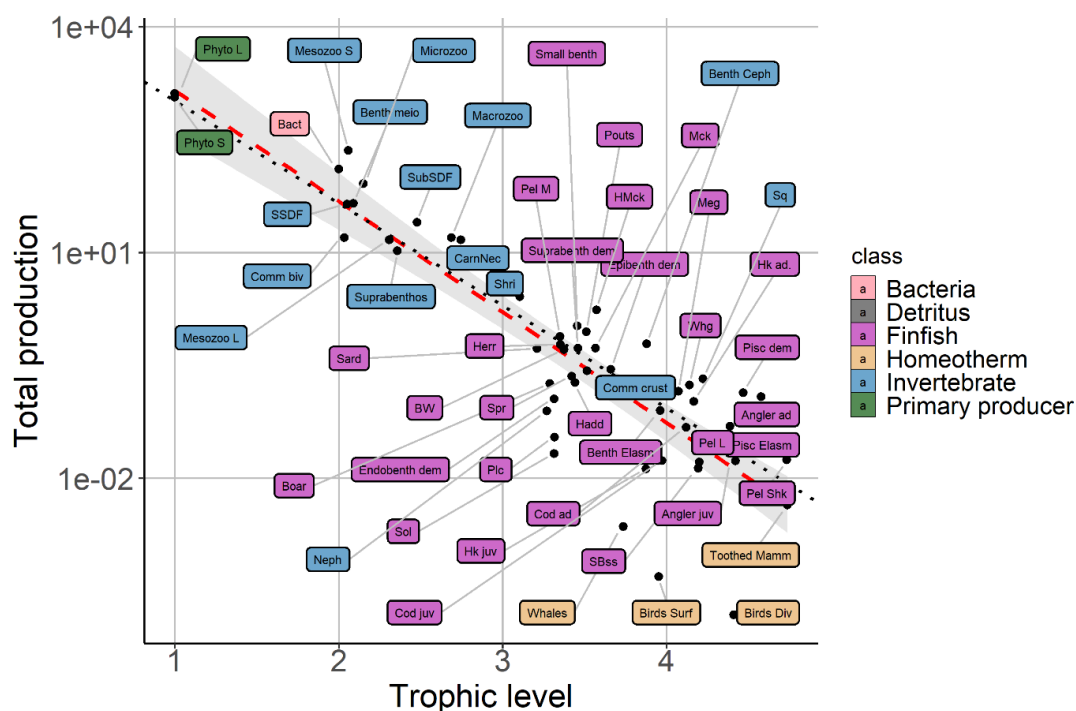


Figure B7 - Negative relationship between the total production (log scale) and the trophic level of functional groups in the Celtic Sea Ecopath model. The red line corresponds to the regression including homeotherms and the black one excluding homeotherms.

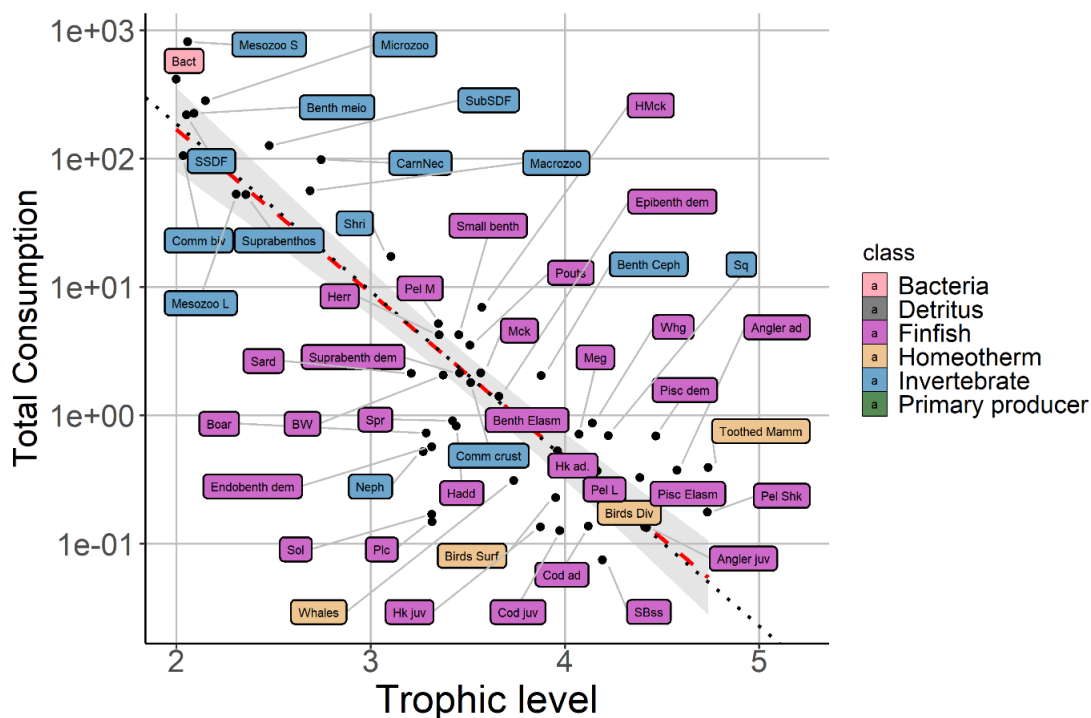


Figure B8 - Negative relationship between the total consumption (log scale) and the trophic level of functional groups in the Celtic Sea Ecopath model. The red line corresponds to the regression including homeotherms and the black one excluding homeotherms.

References SB

- Ainsworth, C., Feriss, B., Leblond, E., & Guénette, S. (2001). The Bay of Biscay, France: 1998 and 1970 models. *Fish. Cent. Res. Rep.*, 9(4), 271–313.
- Agnetta, D., Badalamenti, F., Colloca, F., D’Anna, G., Di Lorenzo, M., Fiorentino, F., et al. (2019). Benthic-pelagic coupling mediates interactions in Mediterranean mixed fisheries: An ecosystem modeling approach. *PLoS ONE* 14, e0210659. doi:10.1371/journal.pone.0210659.
- Bentley, J., Serpetti, N., Fox, C. J., Reid, D. G., and Heymans, J. J. (2019). Modelling the food web in the Irish Sea in the context of a depleted commercial fish community. Part 2: ICES Ecopath with Ecosim Key Run. doi:10.13140/RG.2.2.15136.12809.
- Bentorcha, A., Gascuel, D., and Guénette, S. (2016). Using trophic models to assess the impact of fishing on marine ecosystems. Application to the Bay of Biscay and Celtic Sea case study. *Fisheries Science* Submitted.
- Christensen, V., and Walters, C. J. (2004). Ecopath with Ecosim: methods, capabilities and limitations. *Ecological Modelling* 172, 109–139. doi:10.1016/j.ecolmodel.2003.09.003.
- Day, L., Kopp, D., Robert, M., and Le Bris, H. (2019). Trophic ecology of large gadiforms in the food web of a continental shelf ecosystem. *Progress in Oceanography* 175, 105–114. doi:10.1016/j.pcean.2019.03.007.
- Issac, P., Robert, M., Le Bris, H., Rault, J., Pawlowski, L., and Kopp, D. (2017). Investigating feeding ecology of two anglerfish species, *Lophius piscatorius* and *Lophius budegassa* in the Celtic Sea using gut content and isotopic analyses. *Food Webs* 13, 33–37. doi:10.1016/j.fooweb.2017.08.001.
- Lauria, V. (2012). Impacts of Climate Change and Fisheries on the Celtic Sea Ecosystem (PhD thesis). Plymouth University, School of Science and Technology, Faculty of Marine Science and Engineering, UK. 248 pp.
- Link, J. S. (2010). Adding rigor to ecological network models by evaluating a set of pre-balance diagnostics: A plea for PREBAL. *Ecological Modelling* 221, 1580–1591. doi:10.1016/j.ecolmodel.2010.03.012.
- Morissette, L. (2007). Complexity, cost and quality of ecosystem models and their impact on resilience: a comparative analysis, with emphasis on marine mammals and the Gulf of St. Lawrence (Doctoral dissertation, University of British Columbia).
- Moullec, F., Gascuel, D., Bentorcha, K., Guénette, S., and Robert, M. (2017b). Trophic models: What do we learn about Celtic Sea and Bay of Biscay ecosystems? *Journal of Marine Systems* 172, 104–117. doi:10.1016/j.jmarsys.2017.03.008.
- Théro, H., Rivot, E., Robert, M., Guitton, J., Kopp, D., Gascuel, D., et al. (2020). EcoDiet. R package version 1.0.0.0. doi:10.5281/ZENODO.4081537.

Appendix SC – Spatial-temporal data used to improve low TLs representation

Primary production

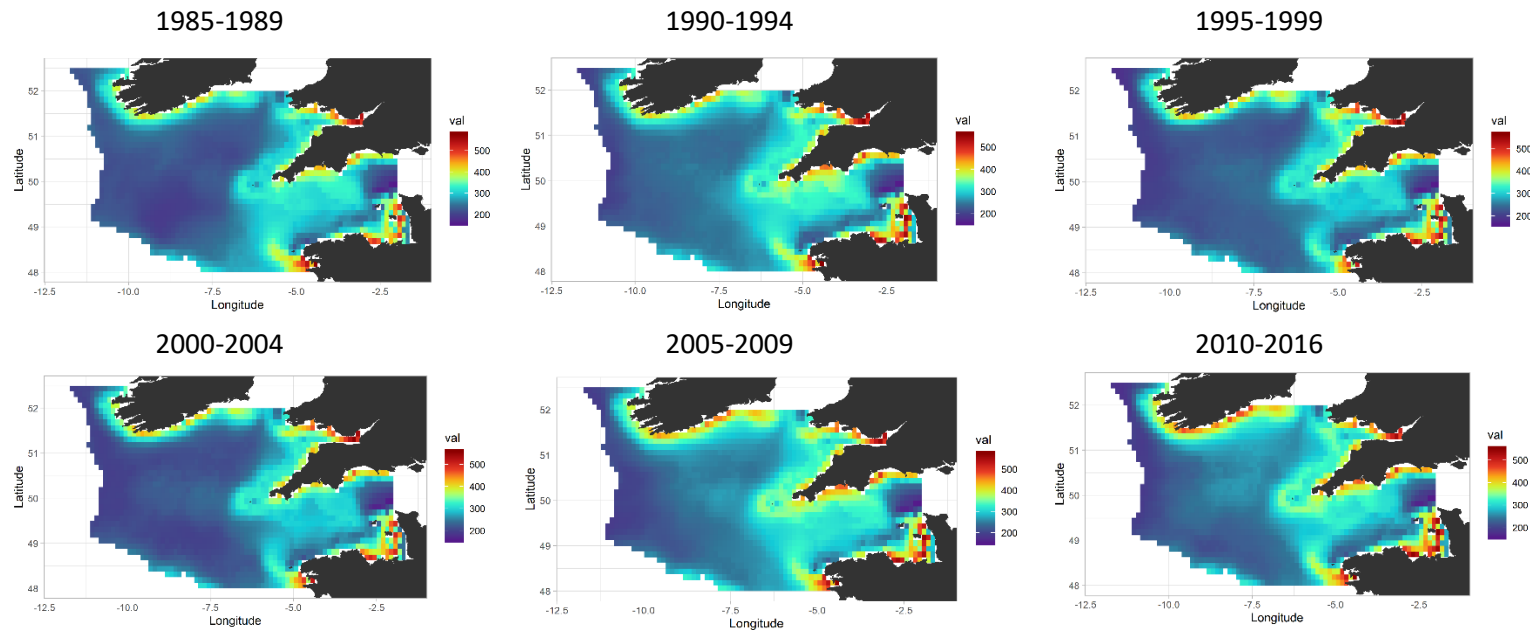
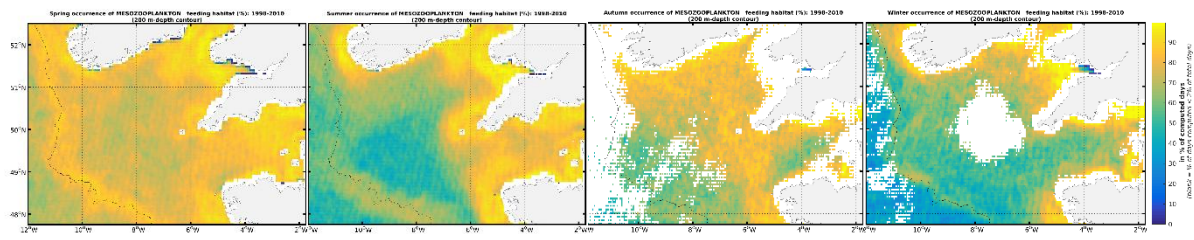


Figure C1 : Mean distribution of the primary production by 5-years periods

Climatologies of the mesozooplankton habitat model

A



B

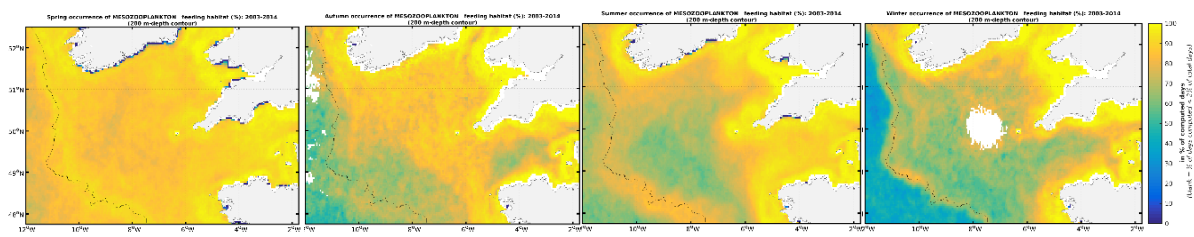


Figure C2 – Mean climatologies of mesozooplankton habitat derived from A/ SeaWiFS sensor (1998-2010) and B/ Modis-Aqua (2002-2013). From left to right: spring, summer, fall, winter. (Druon et al., 2019).

Supplementary Material to: The Celtic Sea Through Time and Space: Ecosystem Modeling to Unravel Fishing and Climate Change Impacts on Food-Web Structure and Dynamics
Hervann P.-Y., Gascuel D., Grüss A., Druon J.-N., Kopp D., Perez I., Piroddi C. and Robert M.

References SC

Druon, J.-N., H  laou  t, P., Beaugrand, G., Fromentin, J.-M., Palialexis, A., and Hoepffner, N. (2019). Satellite-based indicator of zooplankton distribution for global monitoring. *Sci Rep* 9, 4732. doi:10.1038/s41598-019-41212-2.

Appendix SD - Datasets used in statistical habitat models

Fisheries-independent survey database

Description of the surveys considered in this study and of survey data processing

The presence/absence data considered in the present study were collected by European International Bottom Trawl Surveys (IBTS, datras.ices.dk). Because sampling design is highly complex for North Sea IBTS surveys, we chose to not consider the data collected in the North Sea and to use only those IBTS data that were collected on the Western European shelf. The main characteristics of the surveys considered in the present study are described in Tab. D1 and the spatial distribution of the survey data is shown in Fig. D1. One of the main reasons for selecting the surveys detailed in Tab. D1 is that they cover two neighboring Large Marine Ecosystems (i.e. Celtic Biscay Shelf and Iberian Coastal) under similar hydro-climatic influences, while also encompassing a wide range of environmental conditions. Additionally, the widely-ranging species present in the Celtic Sea (e.g. hake *Merluccius merluccius*, horse mackerel *Trachurus trachurus*) are found within the area encompassed by the ensemble of IBTS surveys considered in this study.

IBTS surveys are conducted to monitor demersal fish populations and provide information on commercial species to stock assessments, as well as to gather information on other demersal or benthopelagic species, including cephalopods or crustaceans. Though IBTS surveys are not designed to monitor the abundance of pelagic species, in particular the smallest ones (e.g., sardine, anchovy or sprat), they can provide valuable information on their spatial distributions. At a minimum, the species captured, the number of individuals of each species captured, and their total wet weight are reported for all IBTS hauls. The survey database we constructed thus gathered encounter/non-encounter data aggregated by survey, year, sampling station and functional group. Functional groups include multi-stanza groups since length data were collected in all IBTS surveys for cod, anglerfish, and hake.

Because some of the thirteen surveys collected data using random sampling schemes, while the others employed fixed-station sampling schemes, we summarized the presence/absence data using a 0.125° spatial grid for western European waters before fitting survey data to GAMs. Specifically, for each survey and year: (1) we assigned the encounter/non-encounter data points for a given functional group to their closest 0.125° cell; and (2) assumed the functional group to be encountered within a 0.125° cell if at least one of the data points for the functional group falling in that 0.125° cell was an encounter estimate. The data points falling within a given 0.125° cell were assigned the geographical coordinates of the barycenter of that 0.125° cell.

Supplementary Material to: The Celtic Sea Through Time and Space: Ecosystem Modeling to Unravel Fishing and Climate Change Impacts on Food-Web Structure and Dynamics
Hervann P.-Y., Gascuel D., Grüss A., Druon J.-N., Kopp D., Perez I., Piroddi C. and Robert M.

Table D1 – Characteristics of the fisheries-independent surveys considered in the present study. The surveys conducted in the Celtic Sea are indicated in bold. Datas code refers to the identifier of the survey in the dedicated database of the ICES.

Datras code	Survey name	Used for LTL benthos	Acronym	ICES div.	Months	Gear	Year covered	Number of stations	Reference
–	C ampaigne MAN che O ccidentale pluridisciplinaire	Yes	CAMANOC	7de	Q4	GOV 36/47	2014	2724	Travers-Trolet and Vérin, 2014
FR-CGFS	French Survey in the Eastern English Channel	Yes	FR-CGFS-Q4	7d	Q4	GOV 36/47	1989—2017		Coppin et al., 1988
EVHOE	French Survey in the Celtic Sea and Bay of Biscay	Yes	FR-EVHOE-Q4	7fghj 8ab	Q4	GOV 36/47	Celtic Sea: from 1997 Bay of Biscay: from 1987	4189	Duhamel et al., 2018
IE-IGFS	Irish Survey	No	EI -IGFS-Q4	6a 7bgj	Q4	GOV 36/47	2003-2017	2299	Stokes et al., 2015
NIGFS Q1	Northern Ireland Survey in the Irish Sea	No	UK-NIGFS-Q1	7a	Q1	Rock-hopper	1992—2017	592	Cronne, 2016; Moriarty <i>et al.</i> , 2019
NIGFS Q4	Northern Ireland Groundfish Survey in the Irish Sea	No	UK-NIGFS-Q4	7a	Q4	Rock-hopper	1992—2017	1145	
PT-IBTS	Portuguese Survey in Portuguese Shelf	No	PT-IGFS-Q4	9aMS	Q4	Norwegian Campell Trawl	1992-2017 (\2012)	1144	
SWC-IBTS-Q1	Scottish Western Coast Survey	No	UK-SCO-WCGFS-Q1	6a	Q1	GOV 36/47	1996-2010	450	
SWC-IBTS-Q4	Scottish Western Coast Survey	No	UK-SCO-WCGFS-Q4	6a	Q4	GOV 36/47	1990-2017	331	
SP-NORTH	Spanish Survey in Northern Spanish Shelf	No	SP-NSGFS-Q4	8c 9aN	Q4	BACA 44/60	1990-2017	3086	
SP-PORC	Spanish Survey in the Porcupine Bank	No	SP-PORC-Q3	7bck	Q3	BACA 40/52	2001-2017	1446	
SP-ARSA-Q1	Spanish Survey in the Gulf of Cadiz	No	SP-GCGFS-Q1	9aS	Q1	BACB44/60	1996-2017	571	
SP-ARSA-Q4	Spanish Survey in the Gulf of Cadiz	No	SP-GCGFS-Q4	9aS	Q4	BACB44/60	1997-2017	579	

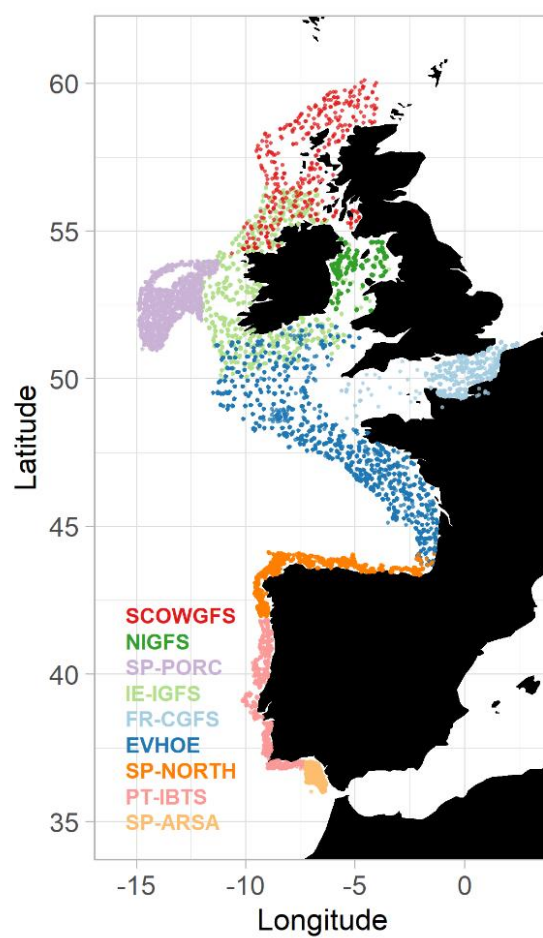


Figure D1 – Spatial distribution of the stations sampled by the International Bottom Groundfish Surveys considered in this study.

Environmental database

Data description and data processing

Topographical, habitat and hydro-climatic variables known to influence fish productivity and spatial distributions in the Celtic Sea were considered in this study. A part of these variables were unvarying through time, namely bathymetry (in m), distance to the coast (in m), and seabed substrate types (categorical). For these variables, unique large scale datasets were available. However, this was not the case for time-varying variables, i.e. surface (first 10 m) and bottom sea temperature, salinity and dissolved oxygen concentration, as well as phytoplankton concentration. Because of the large spatial and temporal extents of the surveys used in the present study, and for the sake of consistency, we aimed to minimize the number of data sources for each environmental variable. Thus, for time-varying variables, we used outputs of two sets of regional physical and biogeochemical models. The first set of models corresponds to the North West Shelf (NWS) reanalyses, analyses, and forecast systems produced by the Met Office (Exeter, UK). It includes a physical model, the Forecasting Ocean Assimilation Model - 7km Atlantic Margin Model (FOAM AMM7), which is coupled to an ecological and sediment model issued from the ERSEM (Baretta *et al.*, 1995). These models were used to provide environmental variables for the largest part of Western Europe since they cover the region spreading from the middle of the Iberian Sea to the North of Scotland. The second set of models corresponds to the Irish Sea – Bay of Biscay (IBI) reanalysis, analysis and forecast systems produced by Mercator Océan (Toulouse, France). They include one physical model, the Mercator Ocean assimilation SAM2, which is coupled to the biogeochemical model PISCES v2 (Aumont *et al.*, 2015). They were used to provide data for the areas from the Gulf of Cadiz to the middle of the Iberian Sea. Data from IBI were corrected from potential biases with NWS. Biases were also corrected between the different versions of the products (i.e. 1985-2013 data for NWS results from the combination of 1985-2013 and 1992-2013 products). All the variables gathered in the environmental database were available at a spatial resolution finer than the 0.125° resolution of the spatial grid for Western European waters. Thus, it was straightforward to process numerical environmental data so that they matched the resolution of spatial grid for Western European waters. The processing of the seabed map issued from EMODnet was slightly different since seabed type is a categorical variable. We calculated the percentage of each seabed type in each cell of the spatial grid for Western European waters. These computations were necessary as Grüss *et al.* (2018)'s method relies on continuous environmental variables only.

The datasets used in our study and the associated references are listed in the Tab. D2.

Table D2 - Characteristics of the environmental variables considered in the present study.

Code	Variable	Data product	Underlying models	Position water column	Temporal extent	Spatial extent	Treatment/Access
Bath	Depth		–	–	–	World	General Bathymetric Chart of the Oceans https://www.gebco.net/
Dist	Distance to coast	computed from Bathy using the "raster" R package	–	–	–	–	–
pMud	Percentage occupancy of 'mud' substrate	Seabed substrate 1:250 000 – Europe © EMODnet Geology, European Commission, 2016 2008 Mesh EUNIS habitat	–	seabed	–	Europe	European Marine Observation and Data Network products https://www.emodnet-seabedhabitats.eu/
pSand	Percentage occupancy of 'mud' sand			seabed	–		
pCoarse	Percentage occupancy of 'mud' coarse			seabed	–		
pMixed	Percentage occupancy of 'mud' mixed			seabed	–		
pRock	Percentage occupancy of 'mud' rock			seabed	–		
SST	Sea surface temperature	NORTHWESTSHELF_REANALYSIS_PHY_004_009	FOAM	first 10m	1985-2013	40°N, 20°W, 65°N, 13°E	E.U. Copernicus Marine Environment Monitoring Service http://marine.copernicus.eu
SBT	Sea bottom temperature	NORTHWESTSHELF_ANALYSIS_FORECAST_PHYS_004_001 (O'Dea et al., 2012, 2017; King et al., 2018)		bottom layer		40°N, 20°W, 65°N, 13°E	
SSS	Sea surface salinity	IBI_REANALYSIS_PHYS_005_002	FOAM	first 10m	2014-2017	40°N, 20°W, 65°N, 13°E	
SBS	Sea bottom salinity			bottom layer	1992-2013	26°N, 19°W, 56°N, 5°E	
MLD	Mixed layer depth	IBI_ANALYSIS_FORECAST_PHYS_005_001 (Bloom et al., 1996 ; Egbert et al., 2002 ; Large et al., 2004 ; Lyard et al., 2006 ; Madec et al., 2008 ; Sotillo et al., 2007 ; Sotillo et al., 2015 ; Umlauf et al., 2003)	SAM2		2014-2017	26°N, 19°W, 56°N, 5°E	
SO2	Surface concentration in dissolved oxygen	NORTHWESTSHELF_REANALYSIS_BIO_004_011 NORTHWESTSHELF_ANALYSIS_FORECAST_BIO_004_002 (Butenschön et al., 2016) IBI_REANALYSIS_BIO_005_003	FOAM/ ERSEM	first 10m	1985-2011	40°N, 20°W, 65°N, 13°E	
BO2	Bottom concentration in dissolved oxygen			bottom layer	2012-2017	40°N, 20°W, 65°N, 13°E	
Chl	Chlorophyll-a concentration		SAM2 / PISCES	first 30m	1992-2017	26°N, 19°W, 56°N, 5°E	

Variables used for habitat modeling

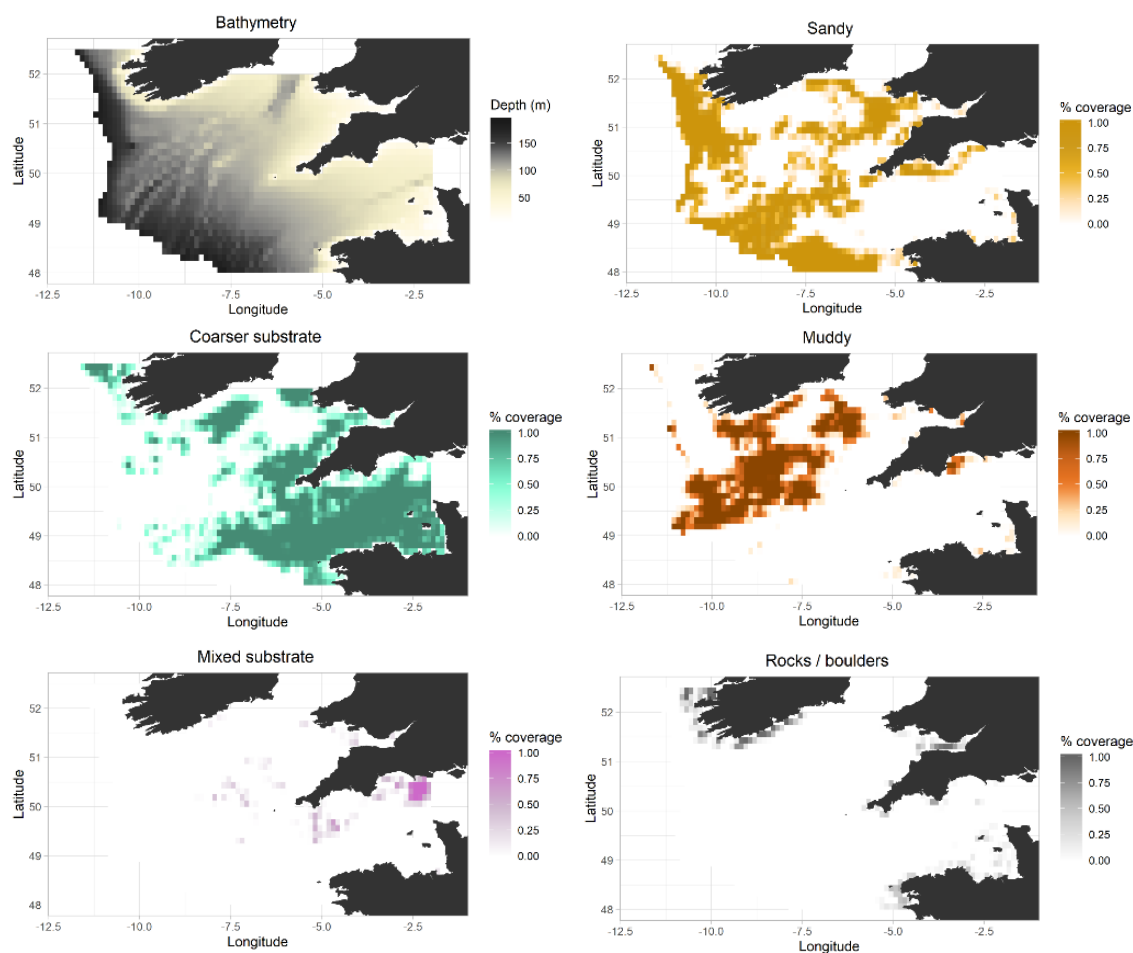
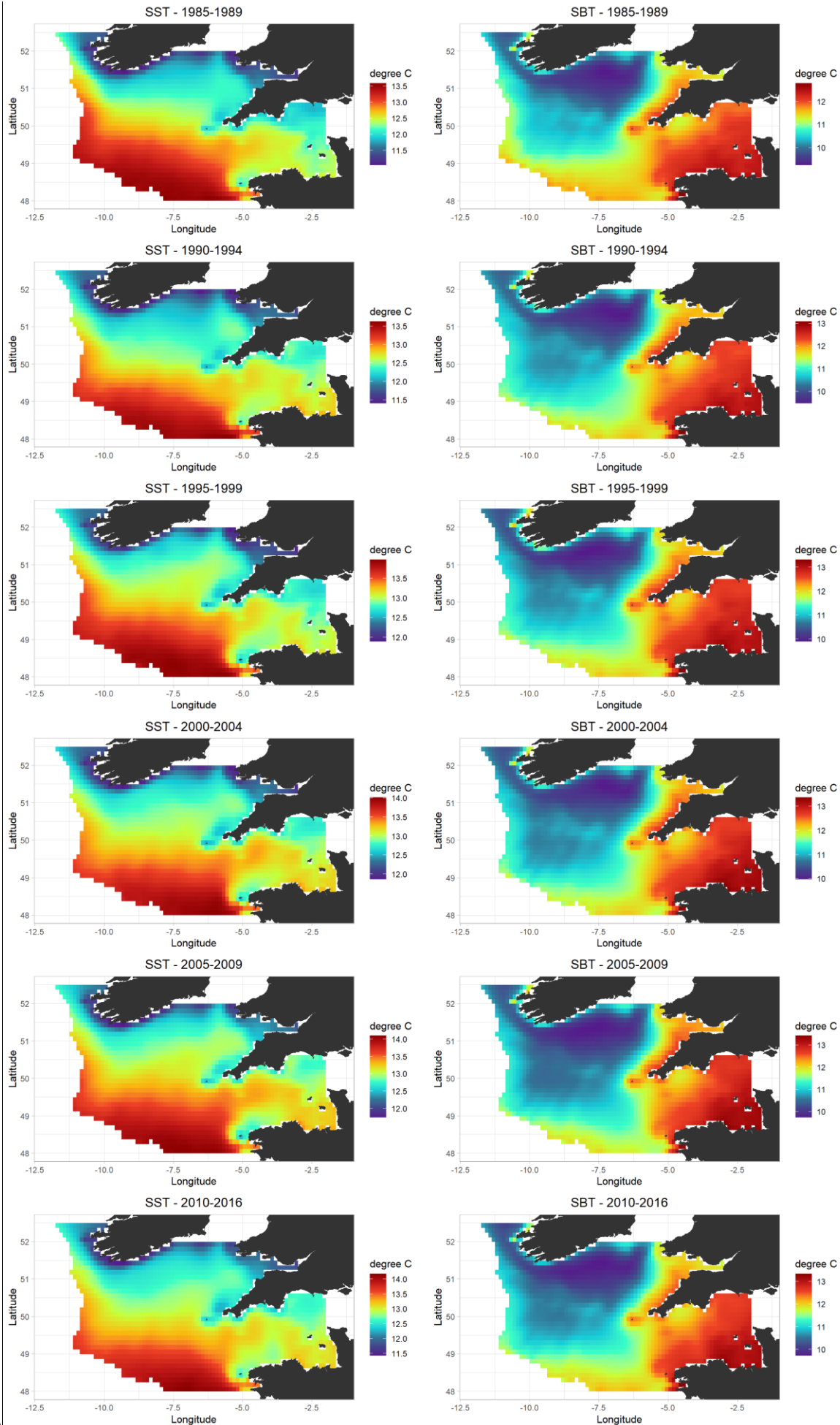
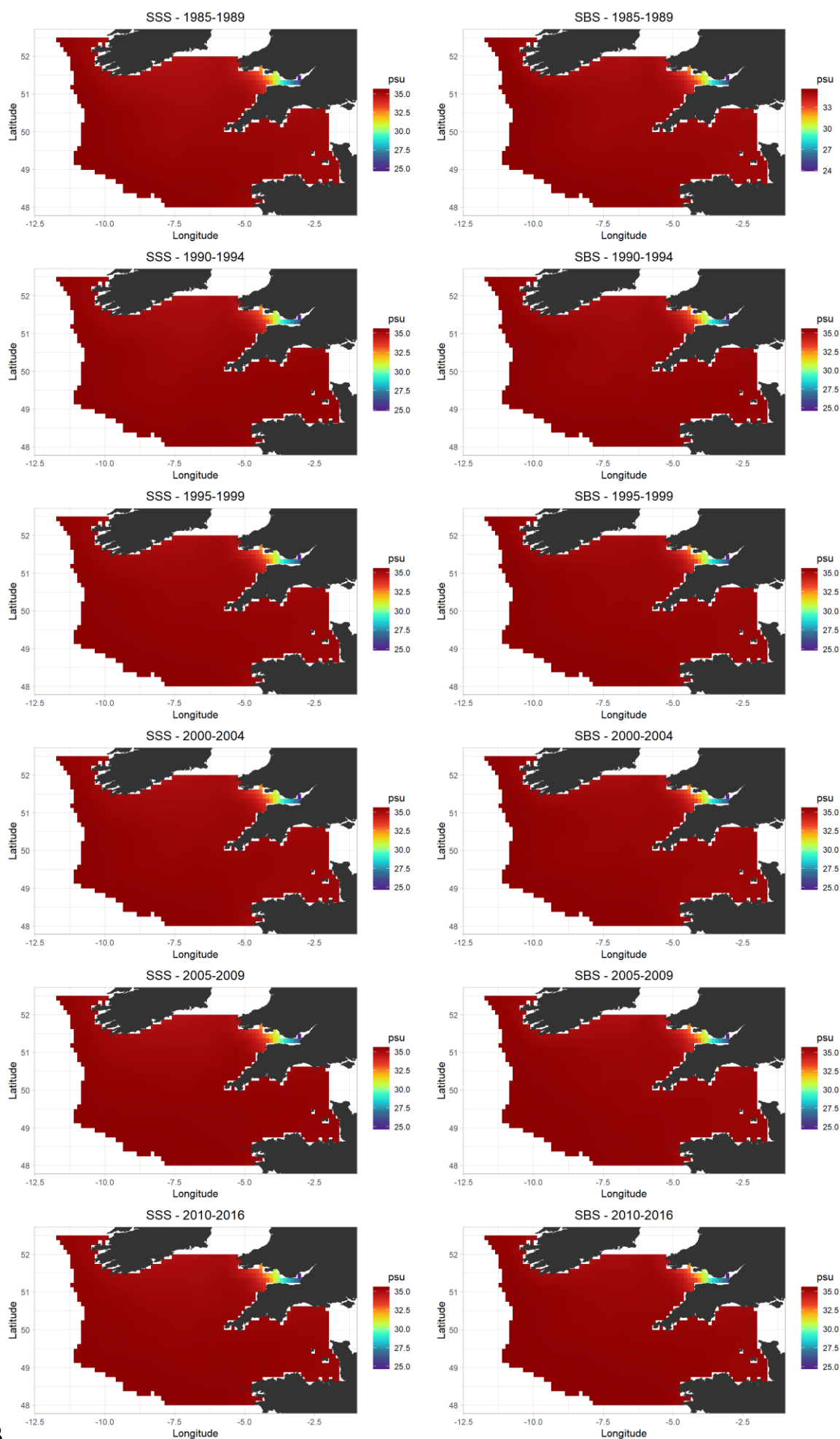
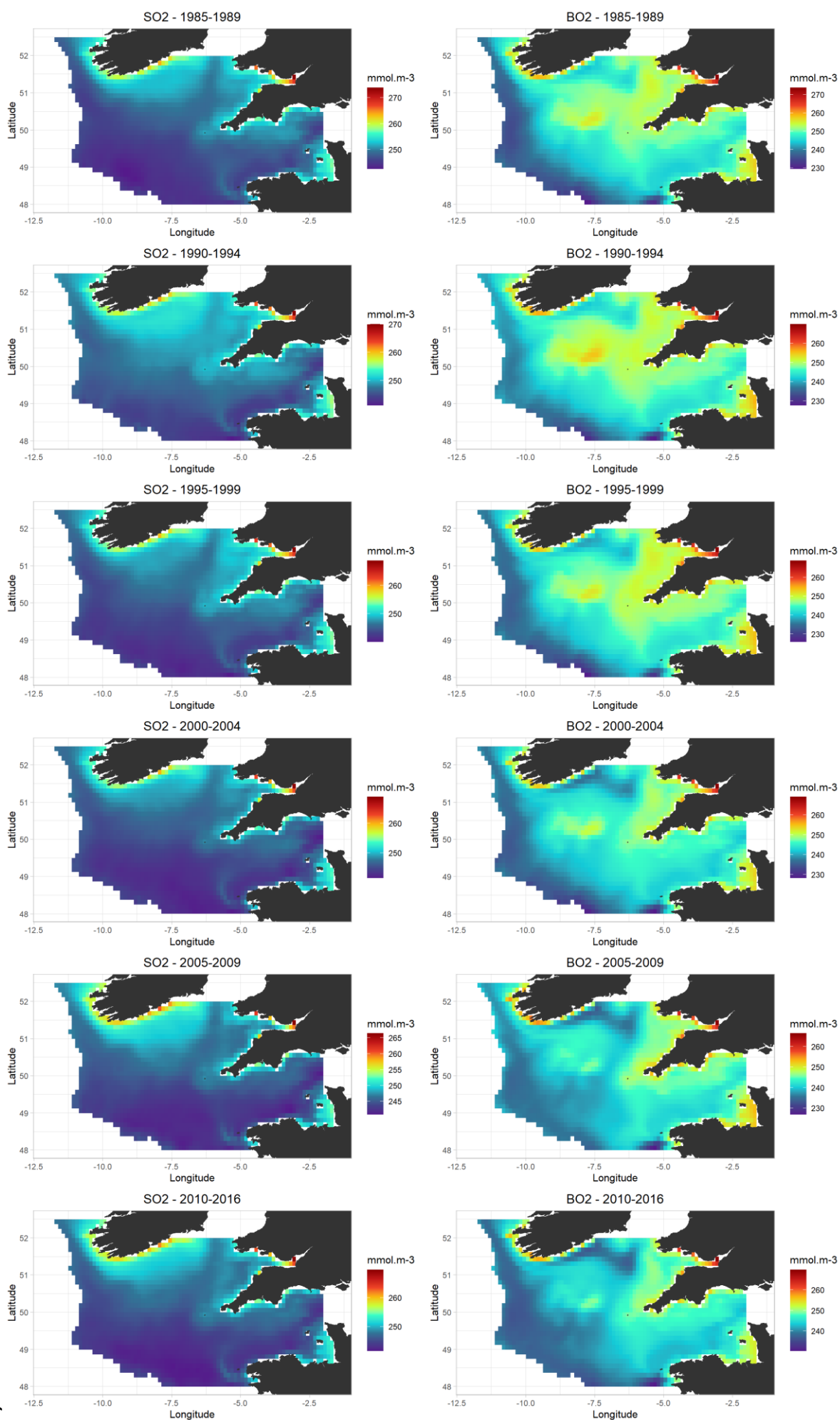


Figure D2: Spatial distribution of time-unvarying environmental variables used for habitat modelling: bathymetry and seabed substrate







Supplementary Material to: The Celtic Sea Through Time and Space: Ecosystem Modeling to Unravel Fishing and Climate Change Impacts on Food-Web Structure and Dynamics
Hervann P.-Y., Gascuel D., Grüss A., Druon J.-N., Kopp D., Perez I., Piroddi C. and Robert M.

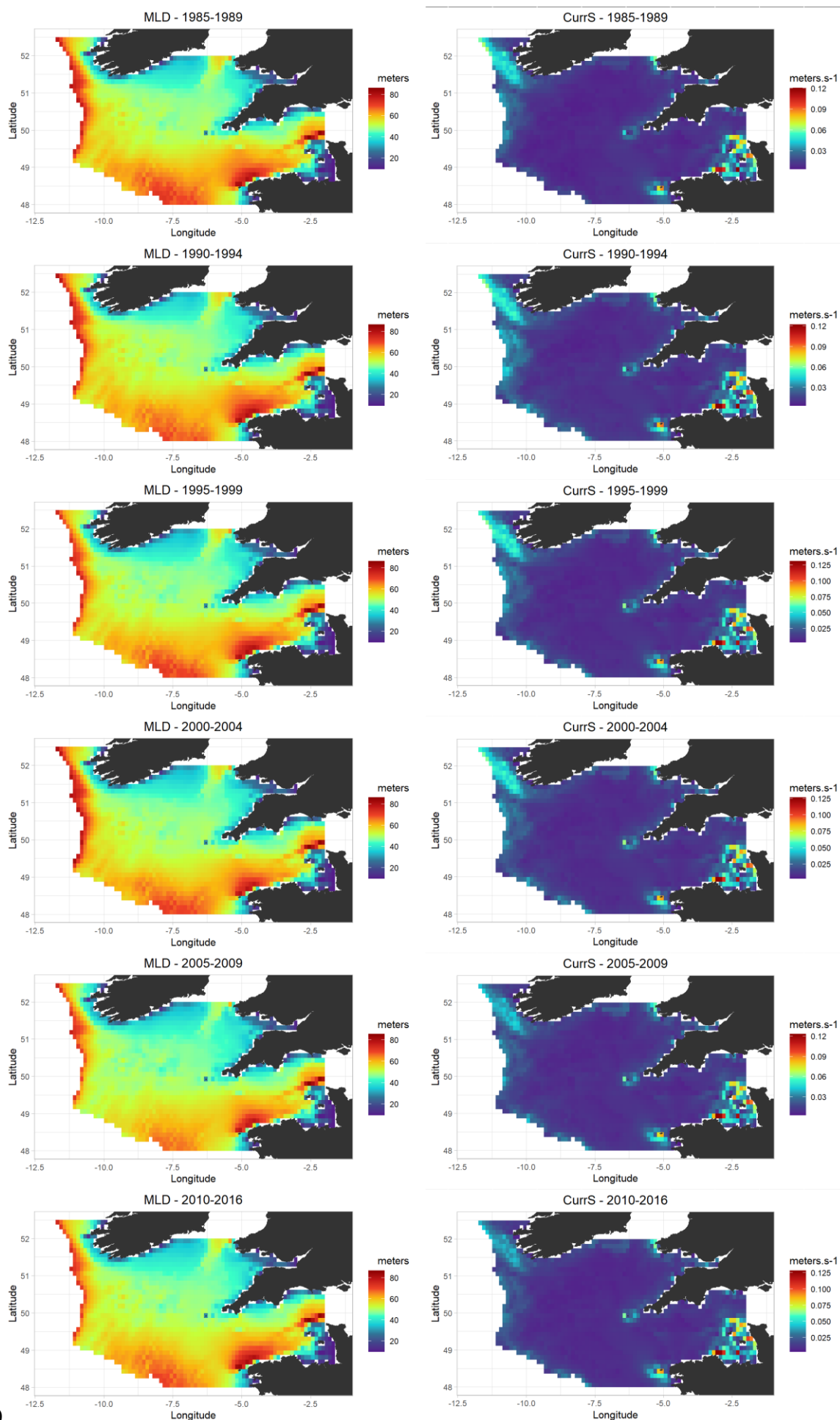


Figure D3 : Mean distribution by 5-years periods of time varying environmental variables used in the habitat modeling framework. SST = sea surface temperature, SBT = sea bottom temperature, SSS= sea surface salinity, SBS= sea bottom salinity, SO₂= dissolved oxygen concentration at the surface, BO₂= dissolved oxygen concentration at the bottom, MLD = mixed layer depth, CurrS = speed of the current at the bottom.

References SD

- Aumont, O., Ethé, C., Tagliabue, A., Bopp, L., and Gehlen, M. (2015). PISCES-v2: an ocean biogeochemical model for carbon and ecosystem studies. *Geoscientific Model Development Discussions* 8, 1375–1509. doi:10.5194/gmdd-8-1375-2015.
- Baretta, J. W., Ebenhöf, W., and Ruudij, P. (1995). The European regional seas ecosystem model, a complex marine ecosystem model. *Netherlands Journal of Sea Research* 33, 233–246. doi:10.1016/0077-7579(95)90047-0.
- Bloom, S. C., Takacs, L. L., Da Silva, A. M., & Ledvina, D. (1996). Data assimilation using incremental analysis updates. *Monthly Weather Review*, 124(6), 1256-1271.
- Butenschön, M., Clark, J., Aldridge, J. N., Allen, J. I., Artioli, Y., Blackford, J., et al. (2016). ERSEM 15.06: a generic model for marine biogeochemistry and the ecosystem dynamics of the lower trophic levels. *Geosci. Model Dev.* 9, 1293–1339. doi:10.5194/gmd-9-1293-2016.
- Coppin, F., Giraldo, C., and Travers-Trolet, M. (1988). CGFS : Channel Ground Fish Survey. doi:10.18142/11.
- Cronne, L. (2016). SISP 15 - manual of the IBTS North Eastern Atlantic surveys. Ser. ICES Surv. Protoc. SISP 15:92. doi: 10.17895/ices.pub.3519
- Duhamel, E., Pawlowski, L., and Garren, F. (2018). EVHOE 2018 cruise, Thalassa R/V. doi:10.17600/18000518.
- Egbert, G. D., & Erofeeva, S. Y. (2002). Efficient inverse modeling of barotropic ocean tides. *Journal of Atmospheric and Oceanic technology*, 19(2), 183-204.
- King, R. R., While, J., Martin, M. J., Lea, D. J., Lemieux-Dudon, B., Waters, J., et al. (2018). Improving the initialisation of the Met Office operational shelf-seas model. *Ocean Modelling* 130, 1–14. doi:10.1016/j.ocemod.2018.07.004.
- Large, W. G., & Yeager, S. G. (2004). Diurnal to decadal global forcing for ocean and sea-ice models: The data sets and flux climatologies.
- Lyard, F., Lefevre, F., Letellier, T., and Francis, O. (2006). Modelling the global ocean tides: modern insights from FES2004. *Ocean Dynamics* 56, 394–415. doi:10.1007/s10236-006-0086-x.
- Madec, G. (2008). NEMO ocean general circulation model reference manuel. Paris: LODYC/IPSL.
- Moriarty, M., Greenstreet, S. P. R., Rasmussen, J., and de Boois, I. (2019). Assessing the State of Demersal Fish to Address Formal Ecosystem Based Management Needs: Making Fisheries Independent Trawl Survey Data ‘Fit for Purpose.’ *Front. Mar. Sci.* 6, 162. doi:10.3389/fmars.2019.00162.
- O’Dea, E., Furner, R., Wakelin, S., Siddorn, J., While, J., Sykes, P., et al. (2017). The CO5 configuration of the 7 km Atlantic Margin Model: large-scale biases and sensitivity to forcing, physics options and vertical resolution. *Geosci. Model Dev.* 10, 2947–2969. doi:10.5194/gmd-10-2947-2017.
- O’Dea, E. J., Arnold, A. K., Edwards, K. P., Furner, R., Hyder, P., Martin, M. J., et al. (2012). An operational ocean forecast system incorporating NEMO and SST data assimilation for the tidally driven European North-West shelf. *Journal of Operational Oceanography* 5, 3–17. doi:10.1080/1755876X.2012.11020128.
- Sotillo, M. G., Cailleau, S., Lorente, P., Levier, B., Aznar, R., Refray, G., et al. (2015). The MyOcean IBI Ocean Forecast and Reanalysis Systems: operational products and roadmap to the future Copernicus Service. *Journal of Operational Oceanography* 8, 63–79. doi:10.1080/1755876X.2015.1014663.
- Sotillo, M. G., Jordi, A., Ferrer, M. I., Conde, J., Tintoré, J., & Alvarez-Fanjul, E. (2007, January). The ESEOO regional ocean forecasting system. In *The Seventeenth International Offshore and Polar Engineering Conference*. International Society of Offshore and Polar Engineers.
- Stokes, D., Gerritsen, H., O’Hea, B., Moore, S.J. & Dransfeld, L., " Irish Groundfish Survey Cruise Report, Sept. 24th – Dec. 17th, 2014", Marine Institute 2014

Travers-Trolet, M., and Verin, Y. (2014). CAMANOC cruise, Thalassa R/V. doi:10.17600/14001900.

Umlauf, L., and Burchard, H. (2003). A generic length-scale equation for geophysical turbulence models. *Journal of Marine Research* 61, 235–265. doi:10.1357/002224003322005087.

Appendix SE – Statistical habitat modeling framework

Additional details on GAM fitting and evaluation

GAMs predictors' selection

The selection of predictors used to predict the probability of encounter of functional groups was implemented before, during, and after GAMs fitting. The a priori selection of predictors based on the literature aimed to retain only a subset of relevant environmental variables for statistical modeling. For example, bottom temperature and seabed type variables were ignored when we developed the GAMs of small epipelagic species. Conversely, they were usually considered when developing the GAMs of demersal and benthic species. In the case of species for which guild attribution was relatively ambiguous, both bottom and surface variables were considered. Specific variables such as the distance to the shore or the chlorophyll-a concentration were considered in statistical analyses, as proxies of variables identified in the literature as influencing the spatial distribution patterns of Celtic Sea organisms.

Environmental variables are known to co-vary in space, hence potentially influencing regression-like statistical methods (Guisan *et al.*, 2002). Two variables with high degree of co-variation can bring redundant information. For instance, one could expect that depth and bottom temperatures are substantially correlated. For this reason, we conducted a collinearity analysis, consisting of calculating Pearson's correlation coefficients between environmental predictors and between environmental predictors and eastings and northings. When a Pearson's correlation coefficient exceeded 0.7 in absolute value, one of the two environmental predictors under consideration was not included in the GAM (Dormann *et al.*, 2013; Leathwick *et al.*, 2008). After the collinearity analysis was conducted, GAMs including the predictors retained after the collinearity analysis were fitted to presence/absence survey data. Smoothing terms for each environmental predictor were estimated using the maximum likelihood (REML) optimization method and while operating a shrinkage approach that, roughly speaking, penalizes over-fitting (Mannocci *et al.*, 2017; Roberts *et al.*, 2016). In addition, as the smoothing parameter approached zero, an extra penalty was applied to each environmental covariate, allowing for the complete removal of a predictor from a GAM when the smoothing parameter was equal to zero (Marra and Wood, 2011). Finally, after a GAM we fitted, if an environmental predictor p-value was larger than 0.05, we removed the environmental predictor from the GAM and refitted it.

Production of preference functions

Fitted GAMs were utilized to produce groups expressing the preferences of functional groups for specific environmental parameter values. For each environmental variable i integrated in a GAM, predictions were made with the GAM over a vector of values ranging between the minimum and maximum values of environmental variable i in the Celtic Sea, while: (1) setting eastings X and northings Y to their values at the

barycenter of the Celtic Sea; 2); (2) keeping the other environmental variables constant at their mean value from the GAM modeled dataset; and (3) setting the two factors, Survey and Year, at their mode from the GAM modeled dataset.

We denote the j th value of the i th environmental predictor by $x_{i,j}$, where $x_{i,j}$ ranges between the minimum and maximum values of the i th environmental predictor in the Celtic Sea, and we denote by $p_{i,j}$ the probability of encounter resulting from the computations described above when the i th environmental variable is set to $x_{i,j}$. Then, for each individual environmental variable i , where $i \in 1, \dots, n$, preferences $f_{i,j}$ are calculated from $p_{i,j}$ values as follows:

$$f_{i,j} = \frac{p_{i,j} - \min\{p_{1,\min}, \dots, p_{n,\min}\}}{\max\{p_{1,\max}, \dots, p_{n,\max}\} - \min\{p_{1,\min}, \dots, p_{n,\min}\}} \quad (\text{Eq E1})$$

where $p_{1,\min}, \dots, p_{n,\min}$ are the minimum values of probability of encounter calculated for environmental parameters $1, \dots, n$; and $p_{1,\max}, \dots, p_{n,\max}$ are the maximum values of probability of encounter calculated for environmental parameters $1, \dots, n$. The calculation of preferences described in Equation 2 allows them to range between 0 and 1 while accounting for the relative effect of each environmental variable on the probability of encounter. Preference functions estimated in this study are displayed in Fig. E1. The preference function for temperature used in Ecosim was issued of the same GAM as the one used in Ecospace except that it was not transformed by Eq E1 and was directly issued of $p_{i,j}$.

Results of the GAMs fitting and prediction of the spatial habitat foraging capacity

Based on these functional responses generated from the predictions of the GAMs fitted for all functional groups and the spatial distribution of environmental variables at each year, the spatial distribution of habitat foraging capacity can be calculated for each year. Maps of habitat foraging capacity for all functional groups for 1985 and 2016 are provided in Fig. E2.

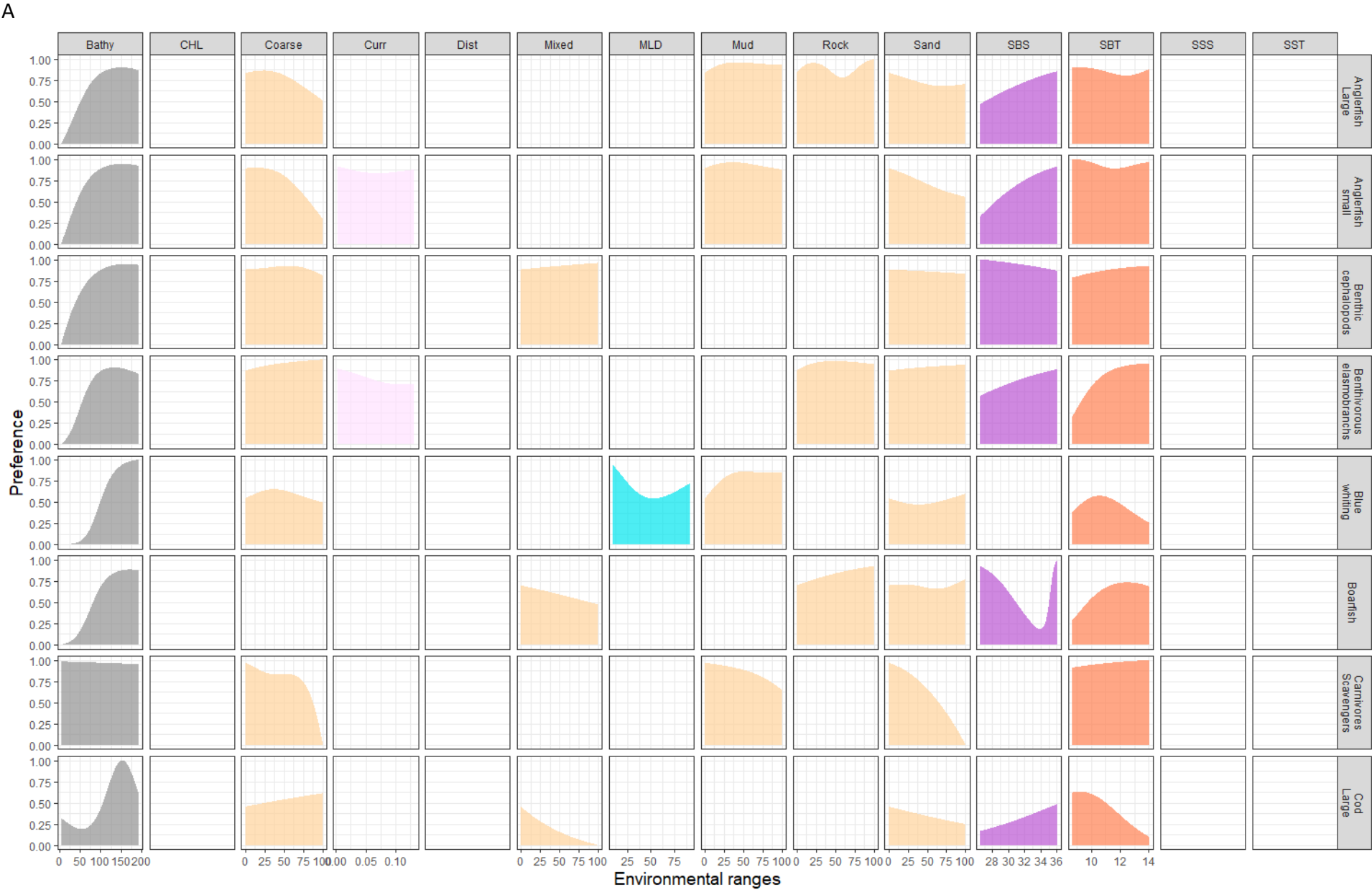
GAM evaluation.

Final models were evaluated using a “Leave Group Out Cross Validation” procedure. In this approach, datasets for a given functional group are split into two sub-datasets (Grüss *et al.*, 2016). The first data subset is composed of a random selection of 60% of the original dataset and is called the “training dataset”. The training dataset is employed to refit a GAM including the final environmental predictors. The second data subset, which comprise the remaining 40% of the data, is used as a “testing dataset” to evaluate/validate the GAM. Two performance metrics are employed within the Leave Group Out Cross Validation procedure: the adjusted coefficient of determination, R^2_{adj} , and area under the receiver operating characteristic curve, AUC. The R^2_{adj} provides an indication of the proportion of the dataset’s variance that is explained by the GAM (Legendre and Legendre, 2012). The AUC quantifies the ability of the GAM to properly separate observed encounters from

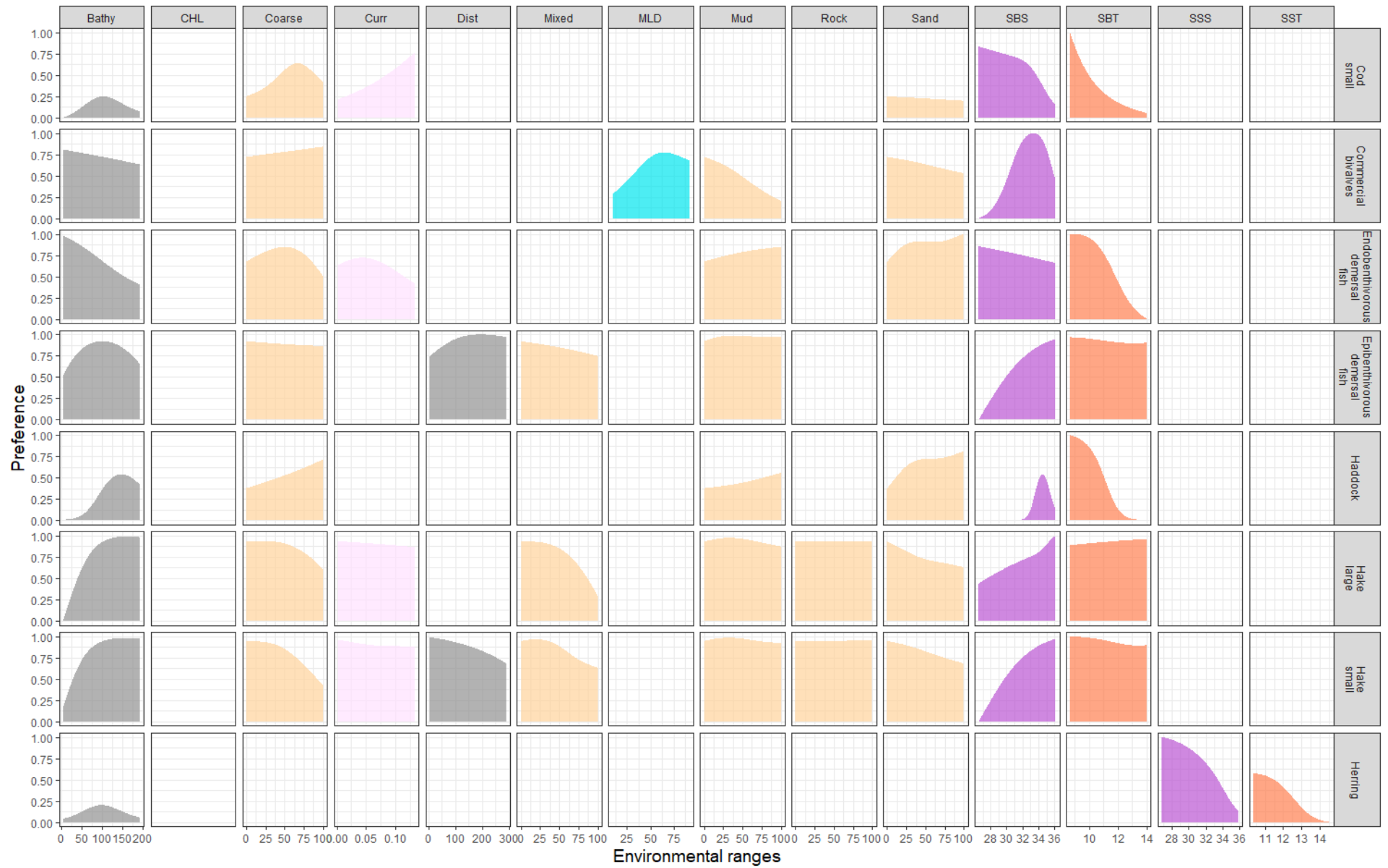
observed non-encounters (Hanley and McNeil, 1982). In other words, the AUC indicates if the model better predicts encounters and non-encounters compared to a random choice. The Leave Group Out Cross Validation procedure was repeated ten times. Thus, for each individual GAM, ten models were trained using distinct training data sets, which were then evaluated using distinct test data sets, each corresponding to a specific training data set. We considered a GAM acceptable when the medians of the ten R^2_{adj} and AUC from the ten iterations were respectively greater than 0.1 and 0.7. The results of the evaluation procedure are presented in Fig. E3.

References SF

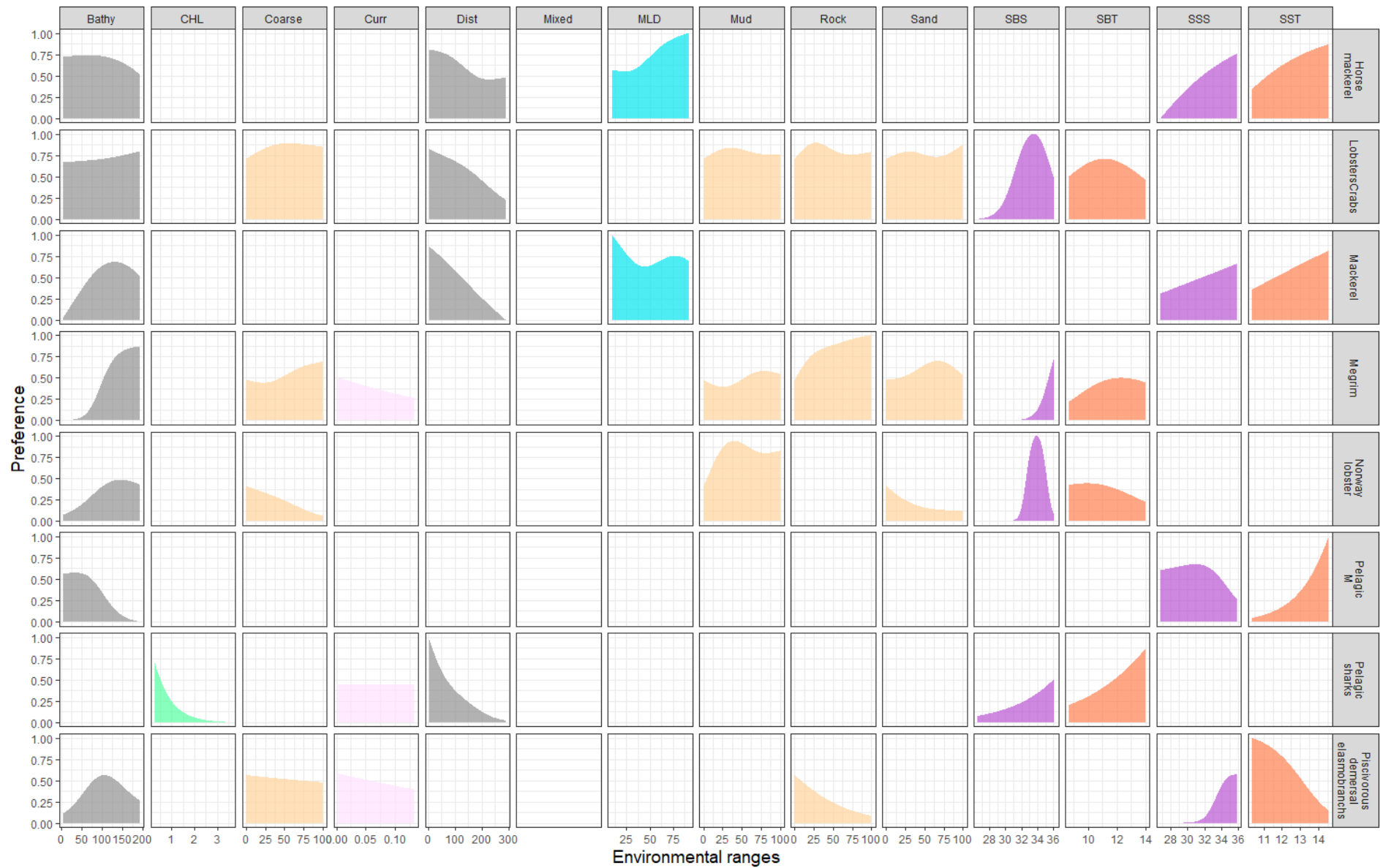
- Dormann, C.F., Elith, J., Bacher, S., Buchmann, C., Carl, G., Carré, G., Marquéz, J.R.G., Gruber, B., Lafourcade, B., Leitão, P.J., Münkemüller, T., McClean, C., Osborne, P.E., Reineking, B., Schröder, B., Skidmore, A.K., Zurell, D., Lautenbach, S., 2013. Collinearity: a review of methods to deal with it and a simulation study evaluating their performance. *Ecography* 36, 27–46. <https://doi.org/10.1111/j.1600-0587.2012.07348.x>
- Grüss, A., Yemane, D., Fairweather, T., 2016. Exploring the spatial distribution patterns of South African Cape hakes using generalised additive models. *African Journal of Marine Science* 38, 395–409. <https://doi.org/10.2989/1814232X.2016.1218367>
- Grüss, A., Drexler, M. D., Ainsworth, C. H., Babcock, E. A., Tarnecki, J. H., and Love, M. S. (2018). Producing Distribution Maps for a Spatially-Explicit Ecosystem Model Using Large Monitoring and Environmental Databases and a Combination of Interpolation and Extrapolation. *Front. Mar. Sci.* 5, 16. doi:10.3389/fmars.2018.00016.
- Grüss, A., Rose, K. A., Justić, D., and Wang, L. (2020). Making the most of available monitoring data: A grid-summarization method to allow for the combined use of monitoring data collected at random and fixed sampling stations. *Fisheries Research* 229, 105623. doi:10.1016/j.fishres.2020.105623.
- Guisan, A., Edwards, T.C., Hastie, T., 2002. Generalized linear and generalized additive models in studies of species distributions: setting the scene. *Ecological Modelling* 157, 89–100. [https://doi.org/10.1016/S0304-3800\(02\)00204-1](https://doi.org/10.1016/S0304-3800(02)00204-1)
- Hanley, J.A., McNeil, B.J., 1982. The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology* 143, 29–36. <https://doi.org/10.1148/radiology.143.1.7063747>
- Leathwick, J., Moilanen, A., Francis, M., Elith, J., Taylor, P., Julian, K., Hastie, T., Duffy, C., 2008. Novel methods for the design and evaluation of marine protected areas in offshore waters: Designing offshore MPAs. *Conservation Letters* 1, 91–102. <https://doi.org/10.1111/j.1755-263X.2008.00012.x>
- Legendre, P., Legendre, L., 2012. Numerical ecology, Third English edition. ed, Developments in environmental modelling. Elsevier, Amsterdam.
- Mannocci, L., Roberts, J.J., Miller, D.L., Halpin, P.N., 2017. Extrapolating cetacean densities to quantitatively assess human impacts on populations in the high seas: Cetacean Densities in the High Seas. *Conservation Biology* 31, 601–614. <https://doi.org/10.1111/cobi.12856>
- Marra, G., Wood, S.N., 2011. Practical variable selection for generalized additive models. *Computational Statistics & Data Analysis* 55, 2372–2387. <https://doi.org/10.1016/j.csda.2011.02.004>
- Roberts, J.J., Best, B.D., Mannocci, L., Fujioka, E., Halpin, P.N., Palka, D.L., Garrison, L.P., Mullin, K.D., Cole, T.V.N., Khan, C.B., McLellan, W.A., Pabst, D.A., Lockhart, G.G., 2016. Habitat-based cetacean density models for the U.S. Atlantic and Gulf of Mexico. *Scientific Reports* 6. <https://doi.org/10.1038/srep22615>



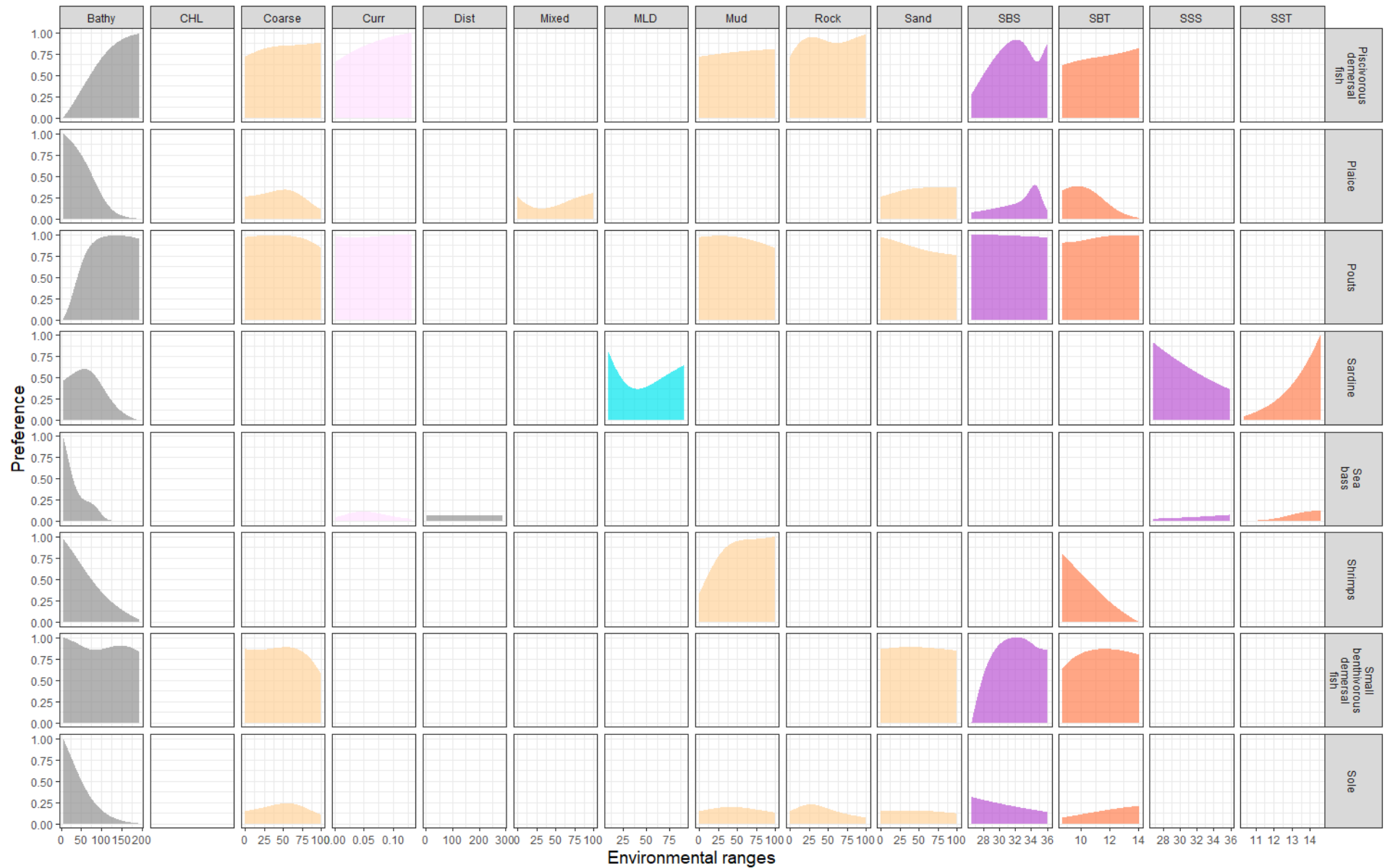
B



C



D



E

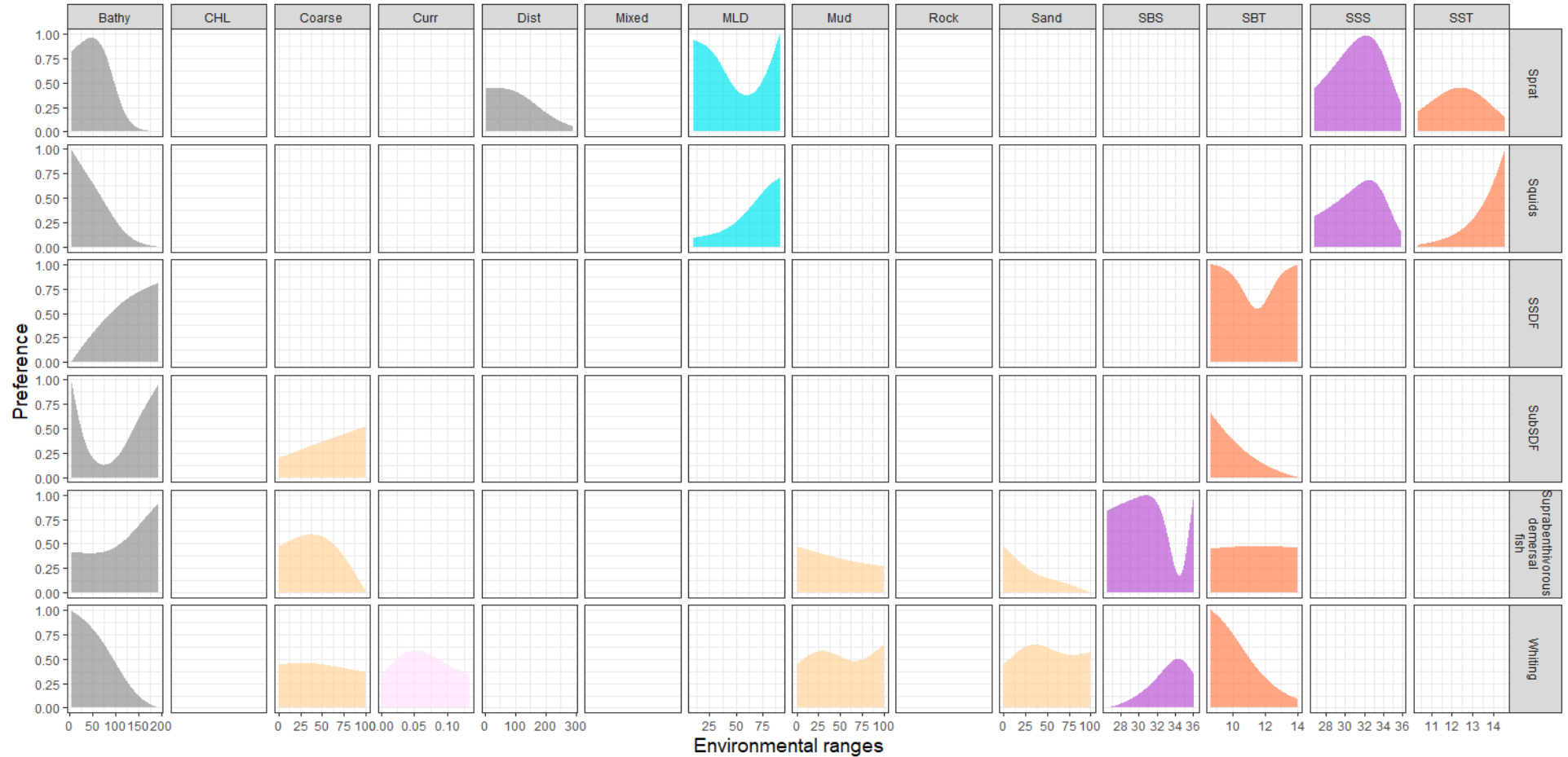


Figure E1 : Preference functions of each Ecopath groups (line) to environmental covariates (columns) for which a habitat model was fitted. Bathy = seabed depth, Curr= speed of the current on the seabed, Dist = distance to the shore, CHL = surface concentration in chlorophyll, MLD = mixed layer depth, SST = sea surface temperature, SBT = sea bottom temperature, SSS= sea surface salinity, SBS= sea bottom salinity, Coarse = proportion of coarseer substrate, Mixed = proportions of mixed sediments substrate, Mud= proportion of muddy substrate, Rock = proportion of rocky substrate, Sand = proportion of sandy substrate. The environmental ranges over which preference functions are represented are defined by the minimum and the maximum values (yearly mean) of the variables observed among all cells of the Ecospace grid and over 1985-2016.

Supplementary Material to: The Celtic Sea Through Time and Space: Ecosystem Modeling to Unravel Fishing and Climate Change Impacts on Food-Web Structure and Dynamics
Hervann P.-Y., Gascuel D., Grüss A., Druon J.-N., Kopp D., Perez I., Piroddi C. and Robert M.

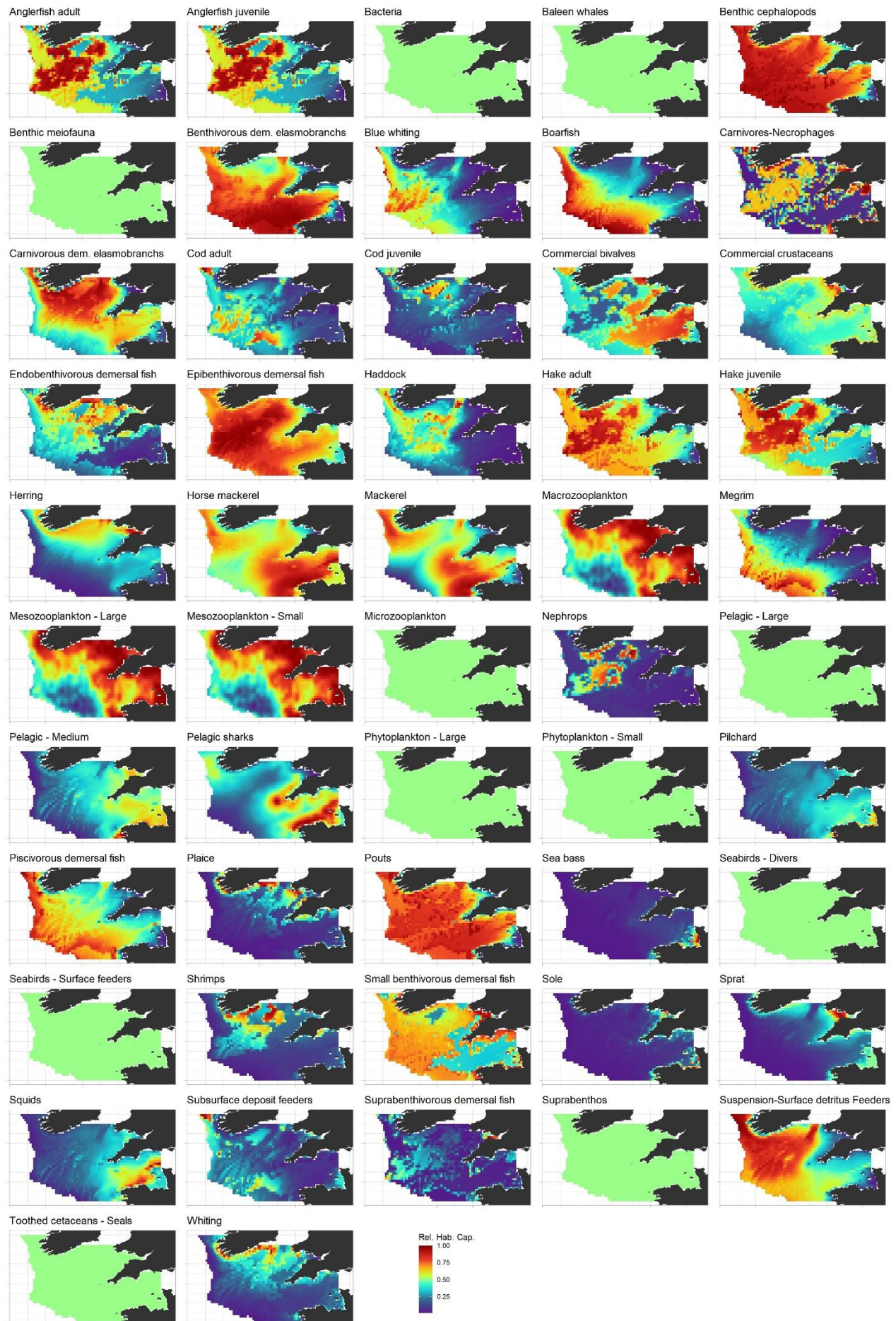


Figure E2 : Relative habitat capacity distribution averaged over the period 1985-1989 for all Celtic Sea Ecospace functional groups. Full green maps correspond to Ecopath groups for which no habitat data has been provided.

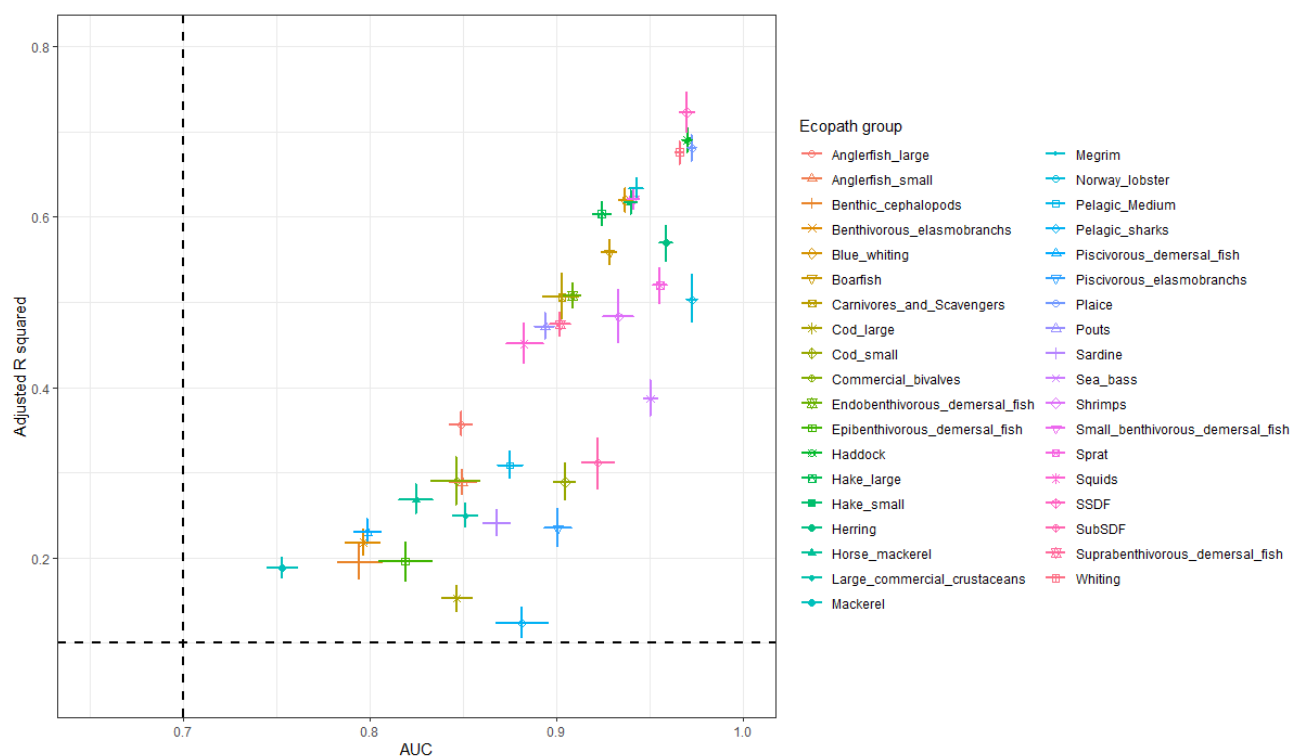


Figure E3 : Quality assessment of the GAMs Performance indices used for GAM evaluation. Adjusted-R2 and AUC respectively quantify GAM fit to the observed data and GAM predictive capacity. The use of a “Leave group out cross-validation” procedure for evaluating GAMs allows for the quantification of uncertainty around the performance indices. The dotted lines represent the lower limits under which a GAM is considered not to have passed the evaluation test.

Appendix SF - The Celtic Sea Ecosim model

A stability test of the Ecopath model prior to any Ecosim simulation

Prior to any further analysis and attempt to forcing and fitting, the Ecopath with Ecosim model was tested for stability. This stability test aimed at identify the problems such as prey-predator cycles over multiple trophic levels, drastic reduction in the biomass of some compartments in response to predation or competition or stock recruitment instabilities (Christensen and Walters, 2000).

This test was performed by modifying the fishing effort time-series so that the system is perturbed and exploring the model's behavior over 100 years of simulation. We tested here 3 types of perturbations related to fishing: a complete cessation of fishing activities from 1985, a brutal and temporary (8 years) increase in fishing effort between 1986 and 1994 and, conversely, a brutal and temporary (8 years too) decrease in fishing effort between 1986 and 1994. The results of the simulations for these three scenarios were compared to the reference one, in which the effort remained constant. The Fig. F1 a), b) and c) represent the ratio between the predicted trends in the alternative scenarios of perturbation and the reference scenario.

The complete cessation of fishing drove a shift toward different status of the ecosystem for which equilibrium was observed around 10 years after the simulations start. The groups with the highest positive variation in biomass were those for which fishing contributed to the largest proportion of the total mortality, including anglerfish, megrim, cod, demersal elasmobranchs, plaice, sole etc. (Fig. F1-a). Note that the compartment with biomass becoming null is the discard detritus box that is not fed anymore when fishing stops. A shorter duration of fishing effort reduction followed by a return to the initial fishing conditions highlighted the reasonable qualitative and quantitative changes in functional groups biomass compared with the reference scenario (Fig. F1-b). In this case, most species responded with an increase in their biomass. Similarly, a short increase in fishing pressure also led to consistent patterns. In this case, the responses among functional groups were more diverse and several invertebrate (e.g.cephalopods) or small prey fish species (e.g. small benthivorous demersal, pouts) due to fishing-induced reduction of their predator abundance.

As a conclusion, the update of the Celtic Sea EwE model did not display particular behavior that required to be corrected. This should be linked to the previous parameterization of the model by Moullec *et al.* (2017) in which Ecosim parameters such as feeding time adjustment rates of switching power were altered from their default value. After this stability test, these Ecosim parameters were also updated using recently published work (Celić *et al.*, 2018). This parameters are presented in the following Tab. F1.

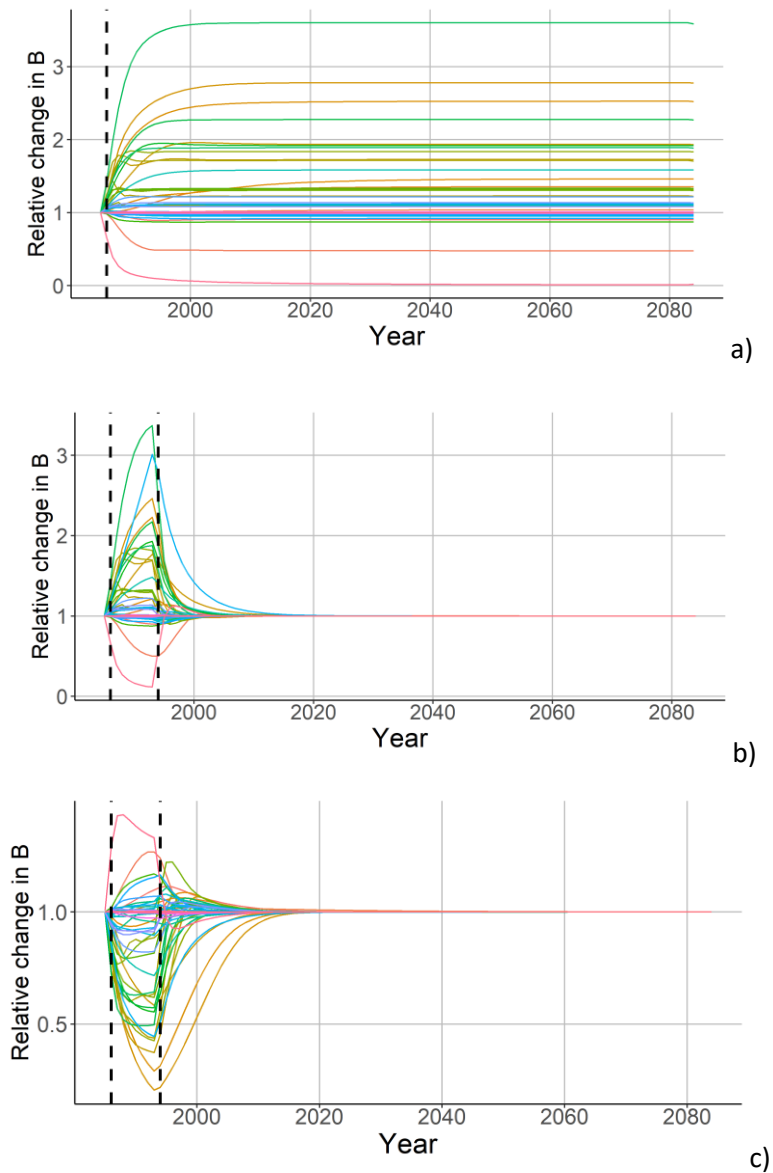


Figure F1 - Relative changes in the biomass trends of each functional group in response to the different fishing-related perturbations: a) the total stop of fishing, b) the temporary fishing effort reduction by a factor 2 c) the temporary fishing effort increase.

More data on abundance trends of multispecies groups

Most of the time-series used in the model developed by Moullec *et al.* (2017) were time-series of catch, for almost all functional groups in the ecosystem, and abundance and fishing mortality time-series for monospecific groups assessed by the ICES. Although abundance indices time-series were included in the Bay of Biscay model in Moullec *et al.* (2017), they were not for the Celtic Sea, the reason being the shortness of the time-series comparatively to the study period. Indeed, the EVHOE survey was only conducted in the Celtic Sea since 1997. Such a late start definitely reduced our ability to explore fishing and environment impacts on the Celtic Sea ecosystem, especially when previous studies have shown that substantial environmental changes occurred during the late 1980s and the mid-1990s. Thus, we looked for other sources of information on abundance trends of fish functional groups, especially the non-assessed multispecific groups. In collaboration with Verena Trenkel (IFREMER) and John Pinnegar (CEFAS), we extracted time-series derived from the UK-WCGFS, an English survey that was conducted in the Celtic Sea between 1982 and 2002. In previous work, the authors have compared the reliability of both EVHOE and UK-WCGFS – derived population abundance indices. Trenkel *et al.* (2004) concluded that there were substantial differences between several indices of population abundance but also concluded that the surveys could provide a good picture of changes operating at the community scale. During the calibration step, each time-series issue from the two surveys was integrated into the model independently (but providing two intermediary short-length relative series of biomass).

Fitting procedure

The Ecosim model was fitted to observed time-series based on a procedure adapted from Bentley *et al.* (2019). This procedure was applied for all the configurations of the model, including or not fishing, plankton-related and temperature forcing. The following consisted into:

- Considering the relative and reference time-series (Table F2) as observed time-series, and forcing the model by various time-series depending on the ecosystem drivers explored (forcing time-series in Table F2 and environmental time-series and preference functions described in the article).
- Following the principle of the automated fitting procedure developed by Scott *et al.* (2016) to determine the best number N of predator-specific vulnerabilities to be estimated while minimizing the AIC score. This step was not conducted using the proper automated fitting procedure as the latter does not allow to specify a primary production anomaly (i.e., it can only be estimated), hence alternative fits were conducted successively by the modeler.
- Setting the N predator-specific vulnerabilities at the values estimated by the best model identified in the above fitting procedure.

- While keeping the N predator-specific estimated values for the vulnerabilities, manually looking at the number N' of complementary predator-prey specific vulnerabilities to be estimated so that the lowest AICc is reached. The number of N' investigated was limited to the maximal number of functional groups with time-series (56) minus N .

The finally selected model for each configuration was the one with N predator-specific vulnerabilities and N' predator-prey –specific vulnerabilities estimated.

Note on the use of zooplankton related time-series:

The mesozooplankton habitat data was used to force the relative changes in the predator-prey dynamics of involving zooplankton as prey and as predator. For trophic interactions between no-zooplankton predator and zooplankton, the ZSH time-series was used to affect the vulnerability of the zooplankton to its predator. This may represent that, when the habitat is suitable for zooplankton, its aggregation in highly productive areas (i.e., fronts) make predation easier for its consumers. For trophic interactions between zooplankton and phytoplankton, the ZSH time-series was used to affect the foraging arena size. This may represent that highly productive areas are characterized by aggregation of both phytoplankton and zooplankton, hence reducing the area over the latter feed on the former. Final, for trophic interactions between two zooplankton groups, the ZSH time-series was used to influence both the vulnerability and the foraging arena size, hence representing the favourable effect on predators and unfavourable effect on prey of their aggregation in the same highly productive areas.

Table F1 – Ecosim inputs – Ecosim group parameters

	<i>Group name</i>	<i>Max rel. P/B</i>	<i>Max rel. feeding time</i>	<i>Feeding time adjust rate</i>	<i>% of oth. mortality sens. to chg. in feeding time</i>	<i>Predator effect on feeding time</i>	<i>Density-dep. catchability</i>	<i>QBmax / QBo</i>	<i>Switching power parameter</i>
1	Seabirds - Divers	2	1.25	0.5	1	0	1	3	2
2	Seabirds - Surface feeders	2	1.25	0.5	1	0	1	3	2
3	Baleen whales	2	1.25	0.1	1	0	1	3	2
4	Toothed cetaceans / Seals	2	1.25	0.1	1	0	1	3	2
5	Pelagic sharks	2	1.25	0.1	1	0	1	10	2
6	Carnivorous dem. elasmobranchs	2	1.25	0.1	1	1	1	3	1
7	Benthivorous dem. elasmobranchs	2	1.25	0.1	1	1	1	3	1
8	Sea bass Anglerfish	2	1.25	0.1	1	1	1	3	2
9	Anglerfish adult	2	1.25	0.1	1	1	1	3	1
10	Anglerfish juvenile Hake	2	1.25	0.2	1	0.75	1	3	1
11	Hake adult	2	1.25	0.1	1	1	1	3	2
12	Hake juvenile Cod	2	1.25	0.2	1	0.75	1	3	1
13	Cod adult	2	1.25	0.1	1	1	1	3	2
14	Cod juvenile	2	1.25	0.2	1	1	1	3	2
15	Whiting	2	1.25	0.1	1	1	1	3	2
16	Haddock	2	1.25	0.1	1	1	1	3	1
17	Pouts	2	1.25	0.1	1	1	1	3	2
18	Megrim	2	1.25	0.1	1	1	1	3	1
19	Sole	2	1.25	0.5	1	1	1	10	0.5
20	Plaice	2	1.25	0.5	1	1	1	10	0.5
21	Piscivorous demersal fish	2	1.25	0.1	1	1	1	3	2
22	Epibenthivorous demersal fish	2	1.25	0.1	1	1	1	3	2
23	Endobenthivorous demersal fish	2	1.25	0.5	1	1	1	10	0.5
24	Suprabenthivorous demersal fish	2	1.25	0.1	1	0.75	1	3	0
25	Small benthivorous demersal fish	2	1.25	0.5	1	0.75	1	3	0.5
26	Mackerel	2	1.25	0.5	1	0.5	1	3	0
27	Horse mackerel	2	1.25	0.5	1	0.5	1	3	0
28	Boarfish	2	1.25	0.5	1	0.5	1	3	0
29	Sprat	2	1.25	0.5	1	0.5	1	3	0
30	Blue whiting	2	1.25	0.5	1	0.5	1	3	0
31	Pilchard	2	1.25	0.5	1	0.5	1	3	0
32	Herring	2	1.25	0.5	1	0.5	1	3	0
33	Pelagic - Large	2	1.25	0.1	1	1	1	10	1
34	Pelagic - Medium	2	1.25	0.5	1	0.5	1	3	0
35	Squids	2	1.25	0.1	1	1	1	3	1
36	Benthic cephalopods	2	1.25	0.1	1	1	1	3	1
37	Commercial crustaceans	2	1.25	0.1	1	0.75	1	3	1
38	Nephrops	2	1.25	0.1	1	0.75	1	3	1
39	Commercial bivalves	2	1	0.05	0	0	1	3	0
40	Shrimps	2	1.25	0.1	1	0.5	1	3	1
41	Carnivores/Necrophages	2	1.25	0.1	1	0.5	1	3	1
42	Suspension/Surface detritus Feeders	2	1.25	0.1	1	0.5	1	3	0
43	Subsurface deposit feeders	2	1.25	0.1	1	0.5	1	3	0
44	Suprabenthos	2	1.5	1	1	0.5	1	3	0
45	Benthic meiofauna	2	1.5	1	0	0	1	3	0
46	Macrozooplankton	2	1.5	1	1	0	1	3	1
47	Mesozooplankton - Large	2	1.5	1	1	0	1	3	0
48	Mesozooplankton - Small	2	1.5	1	1	0	1	3	0
49	Microzooplankton	2	1.5	1	1	0	1	3	0
50	Bacteria	2	1.5	0.05	1	0	1	3	0
51	Phytoplankton - Large	2	1	0.5	1	0	1	1000	0
52	Phytoplankton - Small	2	1	0.5	1	0	1	1000	0

Table F2 – Ecosim inputs – Time-series used for the fit – Contact the authors for data access

<i>Group nb</i>	<i>Group</i>	<i>Data type</i>	<i>Period</i>	<i>Origin</i>
1	Seabirds - Divers	Biomass (forcing)	1985-2016	JNCC
2	Seabirds - Surface feeders	Biomass (forcing)	1985-2016	JNCC
5	Pelagic sharks	Biomass (relative)	1985-1999	UK-WCGFS (CEFAS)
		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
		Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
6	Piscivorous dem. elasmobranchs	Biomass (relative)	1985-1999	UK-WCGFS (CEFAS)
		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
		Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
7	Benthivorous dem. elasmobranchs	Biomass (relative)	1985-1999	UK-WCGFS (CEFAS)
		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
		Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
8	Sea bass	Biomass (relative)	1985-2016	WGCSE
		Catches (reference)	1985-2016	WGCSE - Statlant
		Fishing mortality	1985-2016	WGCSE
9-10	Anglerfish adult / juvenile	Biomass (relative)	1985-2016	WGBIE
		Catches (reference)	1985-2016	WGBIE - Statlant
		Fishing mortality	1985-2016	WGBIE
11-12	Hake adult / juvenile	Biomass (relative)	1985-2016	WGBIE
		Catches (reference)	1985-2016	WGBIE - Statlant
		Fishing mortality	1985-2016	WGBIE
13-14	Cod adult / juvenile	Biomass (relative)	1985-2016	WGCSE
		Catches (reference)	1985-2016	WGCSE - Statlant
		Fishing mortality	1985-2016	WGCSE
15	Whiting	Biomass (relative)	1985-2016	WGCSE
		Catches (reference)	1985-2016	WGCSE - Statlant
		Fishing mortality	1985-2016	WGCSE
16	Haddock	Biomass (relative)	1985-2016	WGCSE
		Catches (reference)	1985-2016	WGCSE - Statlant
		Fishing mortality	1985-2016	WGCSE
17	Pouts	Biomass (relative)	1985-1999	UK-WCGFS (CEFAS)
		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
		Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
18	Megrim	Biomass (relative)	1985-2016	WGBIE
		Catches (reference)	1985-2016	WGBIE - Statlant
		Fishing mortality	1985-2016	WGBIE
19	Sole	Biomass (relative)	1985-2016	WGCSE
		Catches (reference)	1985-2016	WGCSE - Statlant
		Fishing mortality	1985-2016	WGCSE
20	Plaice	Biomass (relative)	1985-2016	WGCSE
		Catches (reference)	1985-2016	WGCSE - Statlant
		Fishing mortality	1985-2016	WGCSE

21	Piscivorous dem. fish	Biomass (relative)	1985-1999	UK-WCGFS (CEFAS)
		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
		Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
22	Epibenthivorous dem. fish	Biomass (relative)	1985-1999	UK-WCGFS (CEFAS)
		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
		Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
23	Endobenthivorous dem. fish	Biomass (relative)	1985-1999	UK-WCGFS (CEFAS)
		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
		Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
24	Suprabenthivorous dem. fish	Biomass (relative)	1985-1999	UK-WCGFS (CEFAS)
		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
		Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
25	Small benthivorous dem. fish	Biomass (relative)	1985-1999	UK-WCGFS (CEFAS)
		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
		Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
26	Mackerel	Biomass (relative)	1985-1999	UK-WCGFS (CEFAS)
		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
		Catches (reference)	1985-2016	Statlant / WGWIDE
		Fishing mortality	1985-2016	WGWIDE
27	Horse mackerel	Biomass (relative)	1985-1999	UK-WCGFS (CEFAS)
		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
		Catches (reference)	1985-2016	Statlant / WGWIDE
		Fishing mortality	1985-2016	WGWIDE
28	Boarfish	Biomass (relative)	1985-1999	UK-WCGFS (CEFAS)
		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
		Catches (reference)	1985-2016	Statlant / WGWIDE
		Fishing mortality	1985-2016	WGWIDE
29	Sprat	Biomass (relative)	1985-2016	Abundance index - HAWG
		Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
30	Blue whiting	Biomass (relative)	1985-1999	UK-WCGFS (CEFAS)
		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
		Catches (reference)	1985-2016	Statlant / WGWIDE
		Fishing mortality	1985-2016	WGWIDE
31	Pilchard	Biomass (relative)	1985-2016	WGHANSA
		Catches (reference)	1985-2016	WGHANSA - Statlant
		Fishing mortality	1985-2016	WGHANSA
32	Herring	Biomass (relative)	1985-2016	HAWG
		Catches (reference)	1985-2016	HAWG - Statlant
		Fishing mortality	1985-2016	HAWG
33	Pelagic - Large	Biomass (relative)	1985-2016	Abundance index - IACCT
		Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
34	Pelagic - Medium	Biomass (relative)	1985-2016	UK-WCGFS (CEFAS)
		Biomass (relative)	1985-2016	FR-EVHOE (Duhamel et al., 2018)

Supplementary Material to: The Celtic Sea Through Time and Space: Ecosystem Modeling to Unravel Fishing and Climate Change Impacts on Food-Web Structure and Dynamics
Hernvann P.-Y., Gascuel D., Grüss A., Druon J.-N., Kopp D., Perez I., Piroddi C. and Robert M.

35		Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
		Biomass (relative)	1985-1999	UK-WCGFS (CEFAS)
	Squids	Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
		Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
36		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
	Benthic cephalopods	Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
37		Biomass (relative)	1985-1999	UK-WCGFS (CEFAS)
		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
	Commercial crustaceans	Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
38		Biomass (relative)	1985-2016	WGCSE
		Catches (reference)	1985-2016	WGCSE - Statlant
	Nephrops	Fishing mortality	1985-2016	WGCSE
39		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
	Commercial bivalves	Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
40		Biomass (relative)	1997-2016	FR-EVHOE (Duhamel et al., 2018)
	Shrimps	Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
41		Biomass (relative)	2003-2016	FR-EVHOE (Duhamel et al., 2018)
	Carnivores/Necrophages	Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
42		Biomass (relative)	2003-2016	FR-EVHOE (Duhamel et al., 2018)
	Suspension/Surface detritus Feeders	Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
43		Biomass (relative)	2003-2016	FR-EVHOE (Duhamel et al., 2018)
	Subsurface deposit feeders	Catches (forcing)	1985-2016	Statlant (ices.dk) - Discard rates from Discardless project (discardless.eu)
46		Biomass (relative)	1985-2016	chaetognaths, hyperiids, mysids, sergestids - CPR survey (cprsurvey.org)
47	Macrozooplankton			
47	Mesozooplankton - Large	Biomass (relative)	1985-2016	large copepods and euphausiids - CPR survey (cprsurvey.org)
48	Mesozooplankton - Small	Biomass (relative)	1985-2016	small copepods - CPR survey (cprsurvey.org)
51	Phytoplankton - Large	Biomass (relative)	1985-2016	total diatoms and dinoflagellates - CPR survey (cprsurvey.org)
52	Phytoplankton - Small	Biomass (relative)	1985-2016	PCI index - CPR survey (cprsurvey.org)

Table F3 – Ecosim fitting outputs: Estimated vulnerabilities

Prey \ predator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
1 Seabirds - Divers																									
2 Seabirds - Surface feeders																									
3 Baleen whales																									
4 Toothed cetaceans / Seals																									
5 Pelagic sharks				1.00E+00	1.00E+00																				
6 Carnivorous dem. elasmobranchs				1.00E+00	1.00E+00																				
7 Benthivorous dem. elasmobranchs				1.00E+00		1.34E+00																			
8 Sea bass								1.00E+00																	
9 Anglerfish adult				1.00E+00	1.00E+00				1.00E+00		6.93E+00		1.13E+00		1.00E+00			1.00E+00			1.00E+00				
10 Anglerfish juvenile									1.00E+00		6.93E+00	1.46E+01	1.13E+00		1.00E+00			1.00E+00			1.00E+00				
11 Hake adult				1.00E+00	1.00E+00				1.00E+00		6.93E+00							1.00E+00			1.00E+00				
12 Hake juvenile								1.00E+00		1.37E+00	6.93E+00	1.46E+01	1.13E+00	1.22E+00	1.00E+00			1.00E+00			1.00E+00				
13 Cod adult				1.00E+00	1.00E+00	1.34E+00			1.00E+00						1.00E+00						1.00E+00				
14 Cod juvenile					1.00E+00	1.34E+00			1.00E+00	1.37E+00					1.00E+00			1.00E+00			1.00E+00	1.00E+00			
15 Whiting			1.00E+00	1.00E+00	1.34E+00	3.96E+00	1.00E+00		1.00E+00	1.37E+00	6.93E+00	1.46E+01	1.13E+00	1.22E+00	1.00E+00			1.00E+00			1.00E+00	1.00E+00			
16 Haddock				1.00E+00	1.34E+00	3.96E+00			1.00E+00	1.37E+00			1.13E+00	1.22E+00	1.00E+00			1.00E+00			1.00E+00	1.00E+00			
17 Pouts			1.00E+00	1.00E+00	1.34E+00	3.96E+00	1.00E+00		1.00E+00	1.37E+00	6.93E+00	1.46E+01	1.13E+00	1.22E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00			1.00E+00	1.00E+00			
18 Megrim									1.00E+00	1.37E+00	6.93E+00		1.13E+00	1.22E+00				1.00E+00							
19 Sole								1.00E+00					1.13E+00	1.22E+00											
20 Plaice				1.00E+00		1.34E+00		1.00E+00	1.00E+00				1.13E+00	1.22E+00				1.00E+00			1.00E+00				
21 Piscivorous demersal fish				1.00E+00	1.00E+00			1.00E+00	1.00E+00				1.13E+00								1.00E+00				
22 Epibenthivorous demersal fish	1.00E+00			1.00E+00	1.00E+00	1.34E+00	3.96E+00		1.00E+00	1.37E+00	6.93E+00	1.46E+01	1.13E+00	1.22E+00	1.00E+00		1.00E+00	1.00E+00			1.00E+00				
23 Endobenthivorous demersal fish			1.93E+00		1.00E+00	1.34E+00			1.00E+00	1.00E+00	1.37E+00		1.13E+00	1.22E+00				1.00E+00			1.00E+00		1.00E+00		
24 Suprabenthivorous demersal fish			1.93E+00	1.00E+00			3.96E+00	1.00E+00	1.00E+00	1.37E+00	6.93E+00	1.46E+01	1.13E+00	1.22E+00	1.00E+00	1.00E+00		1.00E+00			1.00E+00			1.00E+00	
25 Small benthivorous demersal fish						1.34E+00	3.96E+00	1.00E+00	1.00E+00	1.37E+00	6.93E+00	1.46E+01	1.13E+00	1.22E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00		3.04E+00	1.00E+00	1.00E+00	1.62E+00		1.00E+00
26 Mackerel	1.00E+00	1.00E+00	1.93E+00	1.00E+00	1.00E+00	1.34E+00	3.96E+00	1.00E+00	1.00E+00	1.37E+00	6.93E+00		1.13E+00	1.22E+00	1.00E+00			1.00E+00			1.00E+00			1.00E+00	
27 Horse mackerel	1.00E+00	1.00E+00		1.00E+00	1.00E+00	1.34E+00	3.96E+00	1.00E+00	1.00E+00		6.93E+00	1.46E+01	1.13E+00		1.00E+00			1.00E+00			1.00E+00			1.00E+00	1.00E+00
28 Boarfish				1.00E+00		1.34E+00	3.96E+00				6.93E+00				1.00E+00			1.00E+00			1.00E+00				
29 Sprat		1.00E+00		1.00E+00	1.00E+00	1.34E+00	3.96E+00	1.00E+00			6.93E+00	1.46E+01	1.13E+00	1.22E+00	1.00E+00	1.00E+00		1.00E+00			1.00E+00				
30 Blue whiting				1.00E+00	1.00E+00	1.34E+00	3.96E+00	1.00E+00	1.00E+00	1.37E+00	6.93E+00	1.46E+01	1.13E+00	1.22E+00	1.00E+00			1.00E+00			1.00E+00				
31 Pilchard	1.00E+00	1.00E+00		1.00E+00	1.00E+00	1.34E+00		1.00E+00	1.00E+00		6.93E+00	1.46E+01			1.00E+00			1.00E+00			1.00E+00			1.00E+00	
32 Herring	1.00E+00	1.00E+00	1.93E+00	1.00E+00	1.00E+00	1.34E+00	3.96E+00		1.00E+00	1.37E+00	6.93E+00		1.13E+00	1.22E+00	1.00E+00	1.00E+00	1.00E+00			1.00E+00	1.00E+00				
33 Pelagic - Large				1.00E+00	1.00E+00	1.34E+00					6.93E+00														
34 Pelagic - Medium	1.00E+00	1.00E+00	1.93E+00	1.00E+00	1.00E+00	1.34E+00	3.96E+00	1.00E+00	1.00E+00	1.37E+00	6.93E+00	1.46E+01	1.13E+00	1.22E+00	1.00E+00	1.00E+00	1.00E+00			3.04E+00		1.00E+00			
35 Squids			1.93E+00	1.00E+00	1.00E+00		3.96E+00	1.00E+00	1.00E+00	1.37E+00	6.93E+00	1.46E+01	1.13E+00	1.22E+00	1.00E+00			1.00E+00			1.00E+00	1.00E+00			
36 Benthic cephalopods			1.93E+00	1.00E+00		1.34E+00	3.96E+00	1.00E+00	1.00E+00	1.37E+00	6.93E+00	1.46E+01	1.13E+00	1.22E+00	1.00E+00	1.00E+00		1.00E+00	1.91E+00		1.00E+00	1.00E+00			
37 Commercial crustaceans						1.34E+00	3.96E+00	1.00E+00					1.13E+00	1.22E+00						3.04E+00	1.00E+00				
38 Nephrops						1.34E+00	3.96E+00		1.00E+00				1.13E+00	1.22E+00	1.00E+00		1.00E+00	1.00E+00			1.00E+00	1.00E+00		1.00E+00	
39 Commercial bivalves													1.13E+00	1.22E+00											
40 Shrimps					1.00E+00	1.34E+00	3.96E+00	1.00E+00	1.00E+00	1.37E+00	6.93E+00	1.46E+01	1.13E+00	1.22E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.91E+00	3.04E+00	1.00E+00	1.00E+00	1.62E+00	1.00E+00	1.00E+00
41 Carnivores/Necrophages						1.34E+00	3.96E+00	1.00E+00	1.00E+00	1.37E+00	6.93E+00	1.46E+01	1.13E+00	1.22E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.91E+00	3.04E+00	1.00E+00	1.00E+00	1.62E+00	1.00E+00	1.00E+00
42 Suspension/Surface detritus Feeders						1.34E+00	3.96E+00	1.00E+00	1.00E+00	1.37E+00	6.93E+00	1.46E+01	1.13E+00	1.22E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.91E+00	3.04E+00		1.00E+00	1.00E+00	1.62E+00	1.00E+00
43 Subsurface deposit feeders							3.96E+00			1.37E+00	6.93E+00		1.13E+00	1.22E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.91E+00	3.04E+00		1.00E+00	1.00E+00	1.62E+00	1.00E+00
44 Suprabenthos							3.96E+00	1.00E+00			6.93E+00	1.46E+01	1.13E+00	1.22E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.91E+00	3.04E+00	1.00E+00	1.00E+00	1.62E+00	1.00E+00	1.00E+00
45 Benthic meiofauna																			1.91E+00						
46 Macrozooplankton		1.00E+00				1.34E+00						1.46E+01	1.13E+00	1.22E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.91E+00	3.04E+00		1.00E+00	1.00E+00	1.62E+00	1.00E+00
47 Mesozooplankton - Large		1.00E+00	1.93E+00				3.96E+00					1.46E+01	1.13E+00	1.22E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.91E+00	3.04E+00		1.00E+00	1.00E+00	1.62E+00	1.00E+00
48 Mesozooplankton - Small		1.00E+00	1.93E+00				3.96E+00					1.46E+01	1.13E+00	1.22E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00				1.00E+00	1.00E+00	1.62E+00	1.00E+00
49 Microzooplankton																									
50 Bacteria																									
51 Phytoplankton - Large																									
52 Phytoplankton - Small																									
53 Discards	1.00E+00	1.00E+00																							
54 Detritus							3.96E+00										1.00E+00								

Supplementary Material to: The Celtic Sea Through Time and Space: Ecosystem Modeling to Unravel Fishing and Climate Change Impacts on Food-Web Structure and Dynamics
Hernvann P.-Y., Gascuel D., Grüss A., Druon J.-N., Kopp D., Perez I., Piroddi C. and Robert M.

Prey \ predator		27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
1	Seabirds - Divers																								
2	Seabirds - Surface feeders																								
3	Baleen whales																								
4	Toothed cetaceans / Seals																								
5	Pelagic sharks							8.29E+00																	
6	Carnivorous dem. elasmobranchs																								
7	Benthivorous dem. elasmobranchs																								
8	Sea bass																								
9	Anglerfish adult																								
10	Anglerfish juvenile									1.00E+00															
11	Hake adult																								
12	Hake juvenile	1.00E+10								1.00E+00	1.00E+00														
13	Cod adult																								
14	Cod juvenile	1.00E+10								1.00E+00															
15	Whiting	1.00E+10																							
16	Haddock																								
17	Pouts							8.29E+00		1.00E+00															
18	Megrim																								
19	Sole																								
20	Plaice																								
21	Piscivorous demersal fish									1.00E+00															
22	Epibenthivorous demersal fish	1.00E+10						8.29E+00		1.00E+00	1.00E+00														
23	Endobenthivorous demersal fish							8.29E+00																	
24	Suprabenthivorous demersal fish	1.00E+10			3.54E+00			8.29E+00		1.00E+00	1.00E+00														
25	Small benthivorous demersal fish	1.00E+10			3.54E+00					1.00E+00	1.00E+00	2.00E+00	1.97E+00												
26	Mackerel							8.29E+00		1.00E+00															
27	Horse mackerel							8.29E+00		1.00E+00															
28	Boarfish							8.29E+00																	
29	Sprat	1.00E+10						8.29E+00		1.00E+00															
30	Blue whiting	1.00E+10			3.54E+00					1.00E+00															
31	Pilchard							8.29E+00																	
32	Herring	1.00E+10						8.29E+00		1.00E+00															
33	Pelagic - Large							8.29E+00																	
34	Pelagic - Medium	1.00E+10			3.54E+00		1.55E+00	8.29E+00		1.00E+00															
35	Squids	1.00E+10			3.54E+00			8.29E+00		1.00E+00	1.00E+00														
36	Benthic cephalopods							8.29E+00		1.00E+00	1.00E+00														
37	Commercial crustaceans										1.00E+00	2.00E+00													
38	Nephrops				3.54E+00					1.00E+00	1.00E+00														
39	Commercial bivalves											2.00E+00				1.00E+10									
40	Shrimps	1.00E+10	2.67E+01	5.76E+01	3.54E+00		1.55E+00	8.29E+00		1.00E+00	1.00E+00	2.00E+00	1.97E+00		5.88E+01										
41	Carnivores/Necrophages	1.00E+10	2.67E+01	5.76E+01	3.54E+00		1.55E+00	8.29E+00	1.00E+00	1.00E+00	1.00E+00	2.00E+00	1.97E+00		5.88E+01	1.00E+10									
42	Suspension/Surface detritus Feeders	1.00E+10	2.67E+01	5.76E+01	3.54E+00	1.00E+00	1.55E+00	8.29E+00	1.00E+00		1.00E+00	2.00E+00	1.97E+00		5.88E+01	1.00E+10									
43	Subsurface deposit feeders		2.67E+01		3.54E+00				1.00E+00			2.00E+00	1.97E+00		5.88E+01	1.00E+10									
44	Suprabenthos	1.00E+10	2.67E+01		3.54E+00				1.00E+00		1.00E+00		1.97E+00		5.88E+01						1.00E+00				
45	Benthic meiofauna											1.97E+00			5.88E+01			1.00E+00	2.04E+00						
46	Macrozooplankton	1.00E+10	2.67E+01	5.76E+01	3.54E+00	1.00E+00	1.55E+00	8.29E+00	1.00E+00	1.00E+00		2.00E+00	1.97E+00		5.88E+01										
47	Mesozooplankton - Large	1.00E+10	2.67E+01	5.76E+01	3.54E+00	1.00E+00	1.55E+00		1.00E+00	1.00E+00					5.88E+01							1.00E+00			
48	Mesozooplankton - Small	1.00E+10	2.67E+01	5.76E+01	3.54E+00	1.00E+00	1.55E+00		1.00E+00	1.00E+00	1.00E+00				5.88E+01				2.04E+00		1.00E+00	1.71E+01	1.00E+10	2.55E+00	
49	Microzooplankton														5.88E+01				2.04E+00						
50	Bacteria																	1.00E+00	1.00E+00	1.00E+00	1.00E+10				
51	Phytoplankton - Large	1.00E+10				1.00E+00						1.97E+00	1.00E+00	5.88E+01	1.00E+10	1.00E+00	1.00E+00	2.04E+00			1.00E+00	1.71E+01	1.00E+10	2.55E+00	
52	Phytoplankton - Small															1.00E+00	1.00E+00	2.04E+00			1.00E+00	1.71E+01	1.00E+10	2.55E+00	1.00E+00
53	Discards									2.00E+00	1.97E+00					1.00E+10									
54	Detritus									2.00E+00	1.97E+00	1.00E+00	5.88E+01	1.00E+10	1.00E+00	1.00E+00	2.04E+00	1.00E+10	1.00E+00	1.00E+00	1.71E+01	1.00E+10	2.55E+00	1.00E+00	

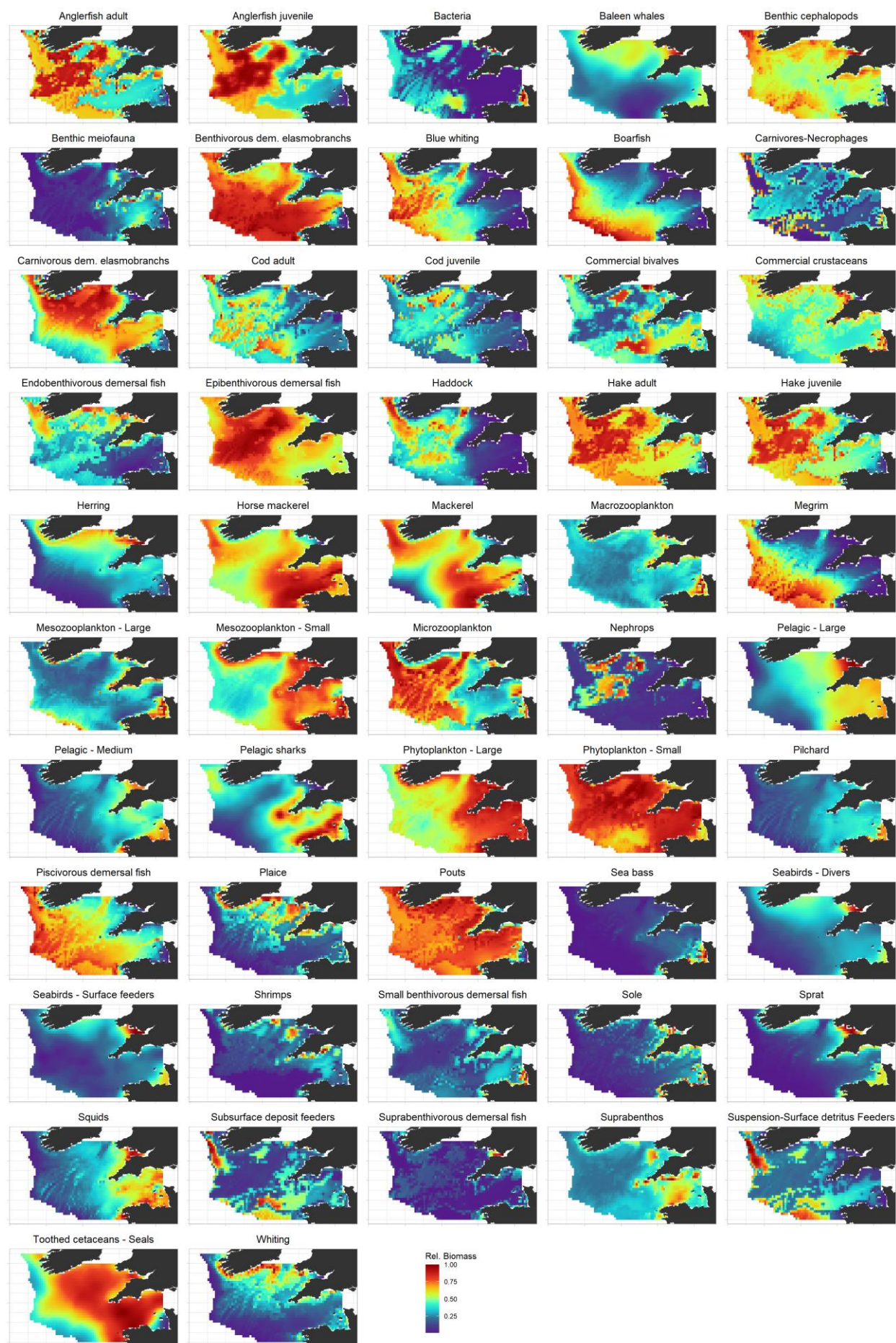
References SF

- Bentley, J., Serpetti, N., Fox, C. J., Reid, D. G., and Heymans, J. J. (2019). Modelling the food web in the Irish Sea in the context of a depleted commercial fish community. Part 2: ICES Ecopath with Ecosim Key Run. doi:10.13140/RG.2.2.15136.12809.
- Celić, I., Libralato, S., Scarcella, G., Raicevich, S., Marčeta, B., and Solidoro, C. (2018). Ecological and economic effects of the landing obligation evaluated using a quantitative ecosystem approach: a Mediterranean case study. *ICES Journal of Marine Science* 75, 1992–2003. doi:10.1093/icesjms/fsy069.
- Christensen, V., and Walters, C. J. (2000). Ecopath with Ecosim: methods, capabilities and limitations.
- Scott, E., Serpetti, N., Steenbeek, J., and Heymans, J. J. (2016). A Stepwise Fitting Procedure for automated fitting of Ecopath with Ecosim models. *SoftwareX* 5, 25–30. doi:10.1016/j.softx.2016.02.002.
- Trenkel, V. M., Pinnegar, J. K., Rochet, M.-J., and Rackham, B. D. (2004). Different surveys provide similar pictures of trends in a marine fish community but not of individual fish populations. *ICES Journal of Marine Science* 61, 351–362. doi:10.1016/j.icesjms.2004.01.004.

Appendix SG - The Celtic Sea Ecospace model

Table G1 – Ecospace input (all years) – Ecospace dispersal table

	Group Name	Base dispersal rate	Rel. disp. in bad hab (%)	Rel. vuln. to pred in bad hab.	Barrier avoid.	Mig. in area movement
1	Seabirds - Divers	275	5	2	0	0.1
2	Seabirds - Surface feeders	275	5	2	0	0.1
3	Baleen whales	652	5	2	0	0.1
4	Toothed cetaceans / Seals	974	5	2	0	0.1
5	Pelagic sharks	275.124	5	2	0	0.1
6	Carnivorous dem. elasmobranchs	275.124	5	2	0	0.1
7	Benthivorous dem. elasmobranchs	157.214	5	2	0	0.1
8	Sea bass	196.517	5	2	0	0.1
9	Anglerfish adult	157.214	5	2	0	0.1
10	Anglerfish juvenile	78.607	5	2	0	0.1
11	Hake adult	196.517	5	2	0	0.1
12	Hake juvenile	78.607	5	2	0	0.1
13	Cod adult	196.517	5	2	0	0.1
14	Cod juvenile	78.607	5	2	0	0.1
15	Whiting	157.214	5	2	0	0.1
16	Haddock	157.214	5	2	0	0.1
17	Pouts	157.214	5	2	0	0.1
18	Megrim	78.607	5	2	0	0.1
19	Sole	75	5	2	0	0.1
20	Plaice	75	5	2	0	0.1
21	Piscivorous demersal fish	157.214	5	2	0	0.1
22	Epibenthivorous demersal fish	115	5	2	0	0.1
23	Endobenthivorous demersal fish	75	5	2	0	0.1
24	Suprabenthivorous demersal fish	78.607	5	2	0	0.1
25	Small benthivorous demersal fish	78.607	5	2	0	0.1
26	Mackerel	235.82	5	2	0	0.1
27	Horse mackerel	235.82	5	2	0	0.1
28	Boarfish	235.82	5	2	0	0.1
29	Sprat	78.607	5	2	0	0.1
30	Blue whiting	157.214	5	2	0	0.1
31	Pilchard	206.81	5	2	0	0.1
32	Herring	206.81	5	2	0	0.1
33	Pelagic - Large	1200	5	2	0	0.1
34	Pelagic - Medium	189.08	5	2	0	0.1
35	Squids	189.08	5	2	0	0.1
36	Benthic cephalopods	189.08	5	2	0	0.1
37	Commercial crustaceans	14.18	1	10	0	0.1
38	Nephrops	543.61	1	10	0	0.1
39	Commercial bivalves	29.871	1	10	0	0.1
40	Shrimps	29.871	1	10	0	0.1
41	Carnivores/Necrophages	29.871	1	10	0	0.1
42	Suspension/Surface detritus Feeders	29.871	1	10	0	0.1
43	Subsurface deposit feeders	29.871	1	10	0	0.1
44	Suprabenthos	29.871	1	10	0	0.1
45	Benthic meiofauna	3	1	10	0	0.1
46	Macrozooplankton	3	5	2	0	0.1
47	Mesozooplankton - Large	3	5	2	0	0.1
48	Mesozooplankton - Small	3	5	2	0	0.1
49	Microzooplankton	3	5	2	0	0.1
50	Bacteria	3	5	2	0	0.1
51	Phytoplankton - Large	3	5	2	0	0.1
52	Phytoplankton - Small	3	5	2	0	0.1
53	Discards	10	5	2	0	0.1
54	Detritus	10	5	2	0	0.1



Supplementary Material to: The Celtic Sea Through Time and Space: Ecosystem Modeling to Unravel Fishing and Climate Change Impacts on Food-Web Structure and Dynamics
Hervann P.-Y., Gascuel D., Grüss A., Druon J.-N., Kopp D., Perez I., Piroddi C. and Robert M.

Figure G1 – Predicted maps of biomass averaged over the 198-1989 period for all Celtic Sea Ecospace

functional groups (t.km⁻²). The color scale represents the relative abundance of the considered group, from zero (blue), to maximum (red).